



RKDF UNIVERSITY RANCHI

MINING ENGINEERING

B.Tech



SYLLABUS (2022-2026) SEMESTER I



RKDF UNIVERSITY RANCHI

B. TECH SEMESTER I

COURSE SCHEME					
BATCH 2021- 2025					
SEMESTER I					
S. No	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Basic Science Course	BSC101	Physics I	100	4
2	Basic Science Course	BSC102	Mathematics I	100	4
3	Engineering Science Course	ESC101	Basic Electrical Engineering	100	4
4	Engineering Science Course	ESC102	Engineering Graphics & Design	100	1
5	Humanities and Social Sciences	HSMC101	English	100	3
PRACTICAL /SESSIONAL					
1	Basic Science Course	BSC101P	Physics I Lab	50	1.5
2	Engineering Science Course	ESC101P	Basic Electrical Engineering Lab	50	1
3	Engineering Science Course	ESC102P	Engineering Graphics & Design Lab	50	2
			TOTAL	650	20.5

Program: B.Tech

Semester: First

Course: Physics I

Course Code: BSC101

Course Objective:

The objective of this course is

- ☛ To understand the students with the concepts of Electromagnetic Theory and applications of Electromagnetic Theory.
- ☛ To understand the theory of Quantum Mechanics.
- ☛ To apply the theory of Mechanics.
- ☛ To illustrate the different Phenomena of Wave Optics.
- ☛ To explain Laser, types of Laser and applications of Laser.
- ☛ To understand the concept of Semiconductor Physics and Semiconductor devices.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO1:** Apply the basic concepts of electromagnetic theory.
- ☛ **CO2:** Explain the dual nature of matter and wave equation.
- ☛ **CO3:** Demonstrate the types of motion and different types of forces.
- ☛ **CO4:** Describes the different Phenomena of Wave Optics, principles of lasers, types of lasers and their application.
- ☛ **CO5:** Design the different types of diodes.

Module 1: Introduction to Electromagnetic Theory (10)

Laplace's and Poisson's equations for electrostatic potential and. Boundary conditions of electric field and electrostatic Potential. Energy of a charge distribution and its expression in terms of electric field. Solving simple electrostatics problems in presence of dielectrics – Point charge at the centre of a dielectric sphere, Bio-Savart law, Divergence and curl of static magnetic field; Magnetization magnetic susceptibility and ferromagnetic, paramagnetic and diamagnetic materials. Faraday's law in terms of EMF produced by changing magnetic flux, Lenz's law. Differential form of Faraday's law Displacement current and Maxwell's equations.

Module II: Quantum Mechanics (7)

Introduction to Quantum mechanics, Wave nature of Particles, Time-dependent and time independent Schrodinger equation for wave function, Uncertainty principle. Solution of stationary-state Schrodinger equation for one dimensional problems– particle in a box, particle in three dimensional box.

Module III: Introduction to Mechanics (8)

Scalars and vectors Forces in Nature; Newton's laws and its completeness in describing particle motion; Form invariance of Newton's Second Law; Solving Newton's equations of motion in polar coordinates; $F = -\text{Grad } V$; Definition and motion of a rigid body in the plane; Rotation in

the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum about a point of a rigid body in planar motion; Euler's laws of motion, their independence from Newton's laws, and their necessity in describing rigid body motion; Examples.

Module IV: Oscillation, Waves and Optics (8)

Mechanical and electrical simple harmonic oscillators. Brewster's Law, total internal reflection, Mirrors & lenses and optical instruments based on them. Amplification of light by population inversion, different types of lasers: gas lasers (He-Ne), solid-state lasers (ruby Laser). Properties of laser beams, applications of lasers in science, engineering and medicine.

Module V: Semiconductor Physics (7)

P-N junction, Metal-semiconductor junction (Ohmic and Schottky); Carrier transport, generation, and recombination; Semiconductor materials of interest for optoelectronic devices. Rates of optical transitions LED: device structure, materials, characteristics, and figures of merit. Types of semiconductor photo detectors -p-n junction, PIN, and Avalanche.

Suggested Reading:

1. *Engineering Physics : R.K.Gaur & S.L. Gupta*
2. *Engineering Physics : G.S. Raghuvanshi*
3. *Modern Engineering Physics: A.S.Vasudeva*
4. *David Griffiths, Introduction to Electrodynamics*
5. *Halliday and Resnick, Physics*
6. *D. J. Griffiths, Quantum*
7. *Engineering Mechanics, 2nd ed. — MK Harbola*
8. *D. J. Griffiths, "Quantum mechanics", Pearson Education.*
9. *S. M. Sze, Semiconductor Devices: Physics and Technology.*



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Program: B.Tech

Semester: First

Course: Physics I

Course Code: BSC101P

List of Laboratory Experiments/Demonstrations:

1. Find the acceleration due to gravity using Kater's pendulum.
2. Find the resistance of a given wire using Meter Bridge.
3. To establish the current voltage relationship for a metallic conductor and find its resistance.
4. To determine the unknown resistance of given wire using Potentiometer.
5. Find the acceleration due to gravity using Simple pendulum.
6. To determine Young's modulus of the material of a given wire using Searle's Apparatus
7. To determine the unknown resistance of given wire using Potentiometer.
8. To determine the wavelength of sodium light by Newton's ring method.
9. To measure the wavelength of Mercury spectrum using Spectrometer grating.
10. To study the variation of magnetic field with distances using Helmholtz Galvanometer.

Program: B.Tech

Semester: First

Course: Mathematics I

Course Code: BSC102

Course Objective:

To familiarize the student with:

- ☛ The mathematical tools needed in evaluating multiple integrals and their usage. .
- ☛ The fallouts of Rolle's Theorem that is fundamental to application of analysis to Engineering problems.
- ☛ The tool of Multivariate Calculus for learning advanced Engineering Mathematics.
- ☛ To deal with functions of multiple integration that are essential in most branches of engineering.
- ☛ The essential tool of sequences and series in a comprehensive manner.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Able to calculate rank of matrix, characteristic equation & characteristic roots & use the applicability of Cayley Hamilton Theorem to find inverse of matrix which is very important in many engineering application
- ☛ CO 2: Ability to understand calculus and its application in engineering
- ☛ CO 3: Gain knowledge about multiple differentiation which is helpful in Engineering & it is also useful in Research & Development
- ☛ CO 4: Gain knowledge about multiple Integration which is helpful in Engineering & it is also useful in Research & Development
- ☛ CO 5: Appreciate knowledge of sequences and series and its application in real world problems.

Module 1: Matrices:

System of linear equations; rank of a matrix, rank-nullity theorem; Symmetric, skew symmetric and orthogonal matrices; Eigen values and eigenvectors; Diagonalization of matrices; Cayley-Hamilton Theorem, and Orthogonal transformation; Cramers rule.

Module 2: Calculus:

Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin theorems with remainders; indeterminate forms and L'Hospital's rule, Evaluation of definite and improper integrals; Beta and Gamma functions and their properties.

Module 3: Multivariable Calculus (Differentiation):

Partial derivatives, directional derivatives, total derivative; Tangent plane and normal line; Maxima, minima and saddle points, Method of Lagrange multipliers.

Module 4: Multivariable Calculus (Integration):

Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes and sphere.

Module 5: Sequences and series:

Convergence of sequence and series, tests for convergence; Power series, Taylor's series, series for exponential, trigonometric and Logarithm functions; Fourier series: Half range sine and cosine series, Parseval's theorem.

Suggested Reading:

1. H.K. Dass, "Advance Engineering Mathematics"; S.Chand & Co., 9th Revised Ed., 2001.
2. B.S. Grewal, "Higher Engineering Mathematics".
3. E. Kreyszig, "Advance Engineering

Program: B.Tech

Semester: First

Course: Basic Electrical Engineering

Course Code: ESC101

Course Objective:

- ☛ To understand and analyze basic Electric and Magnetic circuits.
- ☛ To study the working principles of Electrical Machines and Power Converters.
- ☛ To introduce components of Low Voltage Electrical Installations

Course Outcome:

After the completion of the course, student will be able to:

CO1: Demonstrate and apply basics of electrical engineering networks their applications with different input conditions and analysis.

CO2: Apply fundamentals of electromagnetism and basic design concept of single-phase transformer and measurement of losses.

CO3: Realize the concept and working of different types of DC and AC machines and will be able to select right machine to meet desired needs for implementing engineering solutions.

CO4: Understand the basic circuits of converters, inverters, Switchgear and earthing & types of battery.

Unit 1: D.C. Circuits (8 lectures)

Electrical circuit elements (R, L and C), voltage and current sources, Kirchoff current and voltage laws, analysis of simple circuits with dc excitation. Superposition, Thevenin and Norton Theorems.

Time-domain analysis of first-order RL and RC circuits.

Unit2: A.C. Circuits (8 lectures)

Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor, Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance.

Three-phase balanced circuits, voltage and current relations in star and delta connections.

Unit3: Transformers (6 lectures)

Magnetic materials, BH characteristics, ideal and practical transformer, equivalent circuit, losses in transformers, regulation and efficiency.

Auto-transformer and three-phase transformer connections.

Unit4: Electrical Machines (8 lectures)

Generation of rotating magnetic fields, Construction and working of a three-phase induction motor, Significance of torque-slip characteristic. Loss components and efficiency, starting and speed control of induction motor. Single-phase induction motor.

Construction, working, torque-speed characteristic and speed control of separately excited dc motor. Construction and working of synchronous generators.

Unit 5: Power Converters (6 lectures)

DC-DC buck and boost converters, duty ratio control. Single-phase and three-phase voltage source inverters; sinusoidal modulation.

Unit 6: Electrical Installations (6 lectures)

Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing. Types of Batteries, Important Characteristics for Batteries. Elementary calculations for energy consumption, power factor improvement and battery backup.

Suggested Reading:

1. *Basic Electrical Engineering - D.P. Kothari and I.J. Nagrath, 3rd edition 2010, Tata McGraw Hill.*
2. *Basic Electrical Engineering - D.C. Kulshreshtha, 2009, Tata McGraw Hill.*
3. *Fundamentals of Electrical Engineering, L.S. Bobrow, Oxford University Press, 2011*
4. *Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010*
5. *Electrical Engineering Fundamentals, Vincent Deltoro, Second Edition, Prentice Hall India, 1989*

Program: B.Tech

Semester: First

Course: Basic Electrical Engineering Lab

Course Code: ESC101P

List of Laboratory Experiments/Demonstrations:

1. Basic safety precautions. Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.
2. Measuring the steady-state and transient time-response of R-L, R-C, and R-L-C circuits to a step change in voltage (transient may be observed on a storage oscilloscope). Sinusoidal steady state response of R-L, and R-C circuits – impedance calculation and verification. Observation of phase differences between current and voltage. Resonance in R-L-C circuits.
3. Transformers: Observation of the no-load current waveform on an oscilloscope (non-sinusoidal wave-shape due to B-H curve nonlinearity should be shown along with a discussion about harmonics). Loading of a transformer: measurement of primary and secondary voltages and currents, and power.
4. Three-phase transformers: Star and Delta connections. Voltage and Current relationships (line-line voltage, phase-to-neutral voltage, line and phase currents). Phase-shifts between the primary and secondary side. Cumulative three-phase power in balanced three-phase circuits.
5. Demonstration of cut-out sections of machines: dc machine (commutator-brush arrangement), induction machine (squirrel cage rotor), synchronous machine (field winding - slip ring arrangement) and single-phase induction machine.
6. Torque Speed Characteristic of separately excited dc motor.
7. Synchronous speed of two and four-pole, three-phase induction motors. Direction reversal by change of phase-sequence of connections. Torque-Slip Characteristic of an induction motor. Generator operation of an induction machine driven at super-synchronous speed.
8. Synchronous Machine operating as a generator: stand-alone operation with a load. Control of voltage through field excitation.
9. Demonstration of (a) dc-dc converters (b) dc-ac converters – PWM waveform (c) the use of dc-ac converter for speed control of an induction motor and (d) Components of LT switchgear.

Program: B.Tech

Semester: First

Course: Engineering Graphics & Design

Course Code: ESC102

Course Objective:

- ☛ Introduction to engineering design and its place in society
- ☛ Exposure to the visual aspects of engineering design
- ☛ Exposure to engineering graphics standards
- ☛ Exposure to solid modeling
- ☛ Exposure to computer-aided geometric design
- ☛ Exposure to creating working drawings
- ☛ Exposure to engineering communication

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: To know and understand the conventions and the method of engineering drawing.
- ☛ CO 2: To know the Orthographic & Isometrics projections.
- ☛ CO 3: To know the various projections of regular solids, Sections and Sectional Views of Right Angular Solids.
- ☛ CO 4: To improve visualization skills so that student can apply these skill in developing new products.
- ☛ CO 5: To know about graphical communication and various graphical drawing tools.
- ☛ CO 6: To know the creation of engineering models and their presentation in standard 2D & 3D forms.

Traditional Engineering Graphics: Principles of Engineering Graphics; Orthographic Projection; Descriptive Geometry; Drawing Principles; Isometric Projection; Surface Development; Perspective; Reading a Drawing; Sectional Views; Dimensioning & Tolerances; True Length, Angle; intersection, Shortest Distance.

Computer Graphics: Engineering Graphics Software; -Spatial Transformations; Orthographic Projections; Model Viewing; Co-ordinate Systems; Multi-view Projection; Exploded Assembly; Model Viewing; Animation; Spatial Manipulation; Surface Modelling; Solid Modelling; Introduction to Building Information Modelling (BIM)

Module 1: Introduction to Engineering Drawing covering, Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute; Scales – Plain, Diagonal and Vernier Scales;

Module 2: Orthographic Projections covering, Principles of Orthographic Projections- Conventions - Projections of Points and lines inclined to both planes; Projections of planes inclined Planes - Auxiliary Planes;

Module 3: Projections of Regular Solids covering, those inclined to both the Planes-Auxiliary Views; Draw simple annotation, dimensioning and scale. Floor plans that include: windows, doors, and fixtures such as WC, bath, sink, shower, etc.

Module 4: Sections and Sectional Views of Right Angular Solids covering, Prism, Cylinder, Pyramid, Cone – Auxiliary Views; Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone; Draw the sectional orthographic views of geometrical solids, objects from industry and dwellings (foundation to slab only)

Module 5: Isometric Projections covering, Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple and compound Solids; Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions;

Module 6: Overview of Computer Graphics covering, listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects.; Isometric Views of lines, Planes, Simple and compound Solids];

Module 7: Customisation & CAD Drawing consisting of set up of the drawing page and the printer, including scale settings, Setting up of units and drawing limits; ISO and ANSI standards for coordinate dimensioning and tolerancing; Orthographic constraints, Snap to objects manually and automatically; Producing drawings by using various coordinate input entry methods to draw straight lines, Applying various ways of drawing circles;

Module 8: Annotations, layering & other functions covering applying dimensions to objects, applying annotations to drawings; Setting up and use of Layers, layers to create drawings, Create, edit and use customized layers; Changing line lengths through modifying existing lines (extend/lengthen); Printing documents to paper using the print command; orthographic projection techniques; Drawing sectional views of composite right regular geometric solids and project the true shape of the sectioned surface; Drawing annotation, Computer-aided design (CAD) software modeling of parts and assemblies. Parametric and non-parametric solid, surface, and wireframe models. Part editing and two-dimensional documentation of models. Planar projection theory, including sketching of perspective, isometric, multiview, auxiliary, and section views. Spatial visualization exercises. Dimensioning guidelines, tolerancing techniques; dimensioning and scale multi views of dwelling;

Module 9: Demonstration of a simple team design project that illustrates Geometry and topology of engineered components: creation of engineering models and their presentation in standard 2D blueprint form and as 3D wire-frame and shaded solids; meshed topologies for engineering analysis and tool-path generation for component manufacture; geometric dimensioning and tolerancing; Use of solid-modeling software for creating associative models at the component and assembly levels; floor plans that include: windows, doors, and fixtures such as WC, bath, sink, shower, etc. Applying colour coding according to building drawing practice; Drawing

sectional elevation showing foundation to ceiling; Introduction to Building Information Modelling (BIM).

Suggested Reading:

1. *Engineering Drawing textbook Intro By : N D Bhatt*
2. *Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House*
3. *Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education*
4. *Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication*
5. *Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers*
6. *(Corresponding set of) CAD Software Theory and User Manuals By: R K Dhavan*
7. *Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House*
8. *Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education*
9. *Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication*
10. *Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers*
11. *Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education*
12. *Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication*



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Program: B.Tech

Semester: First

Course: Engineering Graphics & Design Lab

Course Code: ESC102P

Laboratory Experiments/Demonstrations is to be conducted as per module content.

Program: B.Tech

Semester: First

Course: English

Course Code: HSMC101

Course Objective:

The objectives of the course are:

- ☛ To enhance Professional competence in reading, writing, listening and speaking.
- ☛ Switch the approach from providing information about the language to use the language.
- ☛ To use English effectively for study purpose across the curriculum;
- ☛ To revise and reinforce structure already learnt.
- ☛ Introduce Communicative Method of ELT and focusing the teaching pedagogy on the student-centered learning rather than on the teacher-centered learning.
- ☛ Ability to master three major forms of communications which are vital in academic and professional settings namely professional presentations, interviews and group communications respectively.
- ☛ Providing a deep insight into the techniques for delivering effective presentations, winning job interviews, and actively participating in various forms of group communication.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: To design a language component or process to meet desired need within realistic, constraints.
- ☛ CO 2: To analyze the usage of English words in different contexts.
- ☛ CO 3: Develop an understanding of technical and academic articles' comprehension.
- ☛ CO 4: To present oneself at multinational levels knowing the type of different standards of English
- ☛ CO 5: Students will be made to evaluate the correct and error- free writing by being well viewed in rules of and cultivate relevant style of communication & presentation at their work place and also for academic.
- ☛ CO 6: Students will apply it for practical and oral presentation with its presentation skills and voice dynamics.

Detailed contents

1. Vocabulary Building

The concept of Word Formation

Root words from foreign languages and their use in English

Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives.

Synonyms, antonyms, and standard abbreviations.

2. Basic Writing Skills

Sentence Structures

Use of phrases and clauses in sentences

Importance of proper punctuation

Creating coherence

Organizing principles of paragraphs in documents

Techniques for writing precisely

3. Identifying Common Errors in Writing

- Subject-verb agreement
- Noun-pronoun agreement
- Misplaced modifiers
- Articles
- Prepositions
- Redundancies
- Clichés

4. Nature and Style of sensible Writing

- Describing
- Defining
- Classifying
- Providing examples or evidence
- Writing introduction and conclusion

5. Writing Practices

- Comprehension
- Précis Writing
- Essay Writing

6. Oral Communication

(This unit involves interactive practice sessions in Language Lab)

- Listening Comprehension
- Pronunciation, Intonation, Stress and Rhythm
- Common Everyday Situations: Conversations and Dialogues
- Communication at Workplace
- Interviews
- Formal Presentations

Suggested Readings:

1. *Practical English Usage*. Michael Swan. OUP. 1995.
2. *Remedial English Grammar*. F.T. Wood. Macmillan. 2007
3. *On Writing Well*. William Zinsser. Harper Resource Book. 2001
4. *Study Writing*. Liz Hamp-Lyons and Ben Heasley. Cambridge University Press. 2006.
5. *Communication Skills*. Sanjay Kumar and PushpLata. Oxford University Press. 2011.
6. *Exercises in Spoken English*. Parts. I-III. CIEFL, Hyderabad. Oxford University Press

MINING ENGINEERING

B.Tech



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SYLLABUS (2022-2026) SEMESTER II



B.TECH SEMESTER II

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER II					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Basic Science Course	BSC103	Chemistry I	100	4
2	Basic Science Course	BSC104	Mathematics II	100	4
3	Engineering Science Course	ESC103	Programming for Problem Solving	100	3
4	Engineering Science Course	ESC104	Workshop Practice	100	1
5	Mandatory Course	MC101	**Environmental Science	100	0
6	Humanities and Social Sciences	HSMC102	**Life Skills	100	0
PRACTICAL /SESSIONAL					
1	Basic Science Course	BSC103P	Chemistry I Lab	50	1.5
2	Engineering Science Course	ESC103P	Programming for Problem Solving Lab	50	2
3	Engineering Science Course	ESC104P	Workshop Practice Lab	50	2
			TOTAL	750	17.5
** NOTE: Qualifying Non Credit Course					

Program: B.Tech

Semester: Second

Course: Chemistry I

Course Code: BSC103

Course Objective:

- ☛ To bring adaptability to new developments in Engineering Chemistry and to acquire the skills required to become a perfect engineer.
- ☛ To include the importance of water in industrial usage, significance of corrosion control to protect the structures, polymers and their controlled usage.
- ☛ To acquire knowledge of elements, their periodic properties and engineering materials and about fuels and batteries.
- ☛ To acquire required knowledge about few drug molecules and their synthesis and composites.
- ☛ The knowledge gained on polymer chemistry, thermodynamics, spectroscopy, stereochemistry will provide a strong platform to understand the concepts on these subjects for further learning.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Solve the conceptual questions using the knowledge gained by studying the quantum mechanical model of the atom, quantum numbers, electronic configuration, radial and angular distribution curves, shapes of s, p, and d orbitals, and periodicity in atomic radii, ionic radii, ionization energy and electron affinity of elements.
- ☛ **CO 2:** Learn about limitations of classical mechanics and solution in terms of quantum mechanics for atomic/molecular systems. Gain insight into the basic principles of UV, IR and NMR spectroscopic techniques. Use spectroscopic techniques to determine structure and stereochemistry of known and unknown compounds.
- ☛ **CO 3:** Knowledge of ideal gases and real gases, Van der Waals forces, distinguish between intramolecular forces and intermolecular forces
- ☛ **CO 4:** Use the concepts learnt to predict feasibility of chemical reactions and to study the behavior of reactions in equilibrium.
- ☛ **CO 5:** Rationalize periodic properties such as ionization potential, electro negativity, oxidation states.
- ☛ **CO 6:** Identify and differentiate prochirality and chirality at centres, axis, planes and determine the absolute configuration. Evaluate the stability of various conformers of acyclic and cyclic systems using steric, electronic and stereo electronic effects and correlate them to reactivity.
- ☛ **CO 7:** Gain insight into retro-synthesis approach in relation to drug design and drug discovery.

Module I: Atomic and molecular structure (8)

Schrodinger equation. Molecular orbitals of diatomic molecules and plots of the multicentre orbitals. Equations for atomic and molecular orbitals. Energy level diagrams of diatomics. Pi-

molecular orbitals of butadiene and benzene and aromaticity. Crystal field theory and the energy level diagrams for transition metal ions and their magnetic properties. Band structure of solids and the role of doping on band structures.

Module II: Spectroscopic techniques and applications (7 lectures)

Principles of spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging.

Module III: Intermolecular forces and potential energy surfaces (4 lectures)

Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena.

Module IV: Use of free energy in chemical equilibria (6 lectures)

Thermodynamic functions: energy, entropy and free energy. Estimations of entropy and free energies. Free energy and emf. Cell potentials, the Nernst equation and applications. Acid base, oxidation reduction and solubility equilibria. Water chemistry. Corrosion.

Module V: Periodic properties (6 Lectures)

Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries

Module VI: Stereochemistry (5 lectures)

Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds.

Module VII: Organic reactions and synthesis of a drug molecule (4 lectures)

Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule.

Suggested Reading:

- (i) *University chemistry*, by B. H. Mahan
- (ii) *Chemistry: Principles and Applications*, by M. J. Sienko and R. A. Plane
- (iii) *Fundamentals of Molecular Spectroscopy*, by C. N. Banwell
- (iv) *Engineering Chemistry (NPTEL Web-book)*, by B. L. Tembe, Kamaluddin and M. S. Krishnan
- (v) *Physical Chemistry*, by P. W. Atkins
- (vi) *Organic Chemistry: Structure and Function* by K. P. C. Volhardt and N. E. Schore.
- (vii) *Engineering Chemistry* by Jain & Jain.
- (viii) *Engineering Chemistry* by O P Agarwal.

Program: B.Tech

Semester: Second

Course: Chemistry I Lab

Course Code: BSC103P

List of Laboratory Experiments/Demonstrations:

1. Determination of surface tension and viscosity
2. Thin layer chromatography
3. Ion exchange column for removal of hardness of water
4. Determination of chloride content of water
5. Colligative properties using freezing point depression
6. Determination of the rate constant of a reaction
7. Determination of cell constant and conductance of solutions
8. Potentiometry - determination of redox potentials and emfs.
9. Synthesis of a polymer/drug
10. Saponification/acid value of an oil
11. Chemical analysis of a salt
12. Lattice structures and packing of spheres
13. Models of potential energy surfaces
14. Chemical oscillations- Iodine clock reaction
15. Estimation of Hcl solution supplied titrating it against N/10 Hcl solution.
16. Adsorption of acetic acid by charcoal
17. Use of the capillary viscosimeters to demonstrate the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg

Program: B.Tech

Semester: Second

Course: Mathematics II

Course Code: BSC104

Course Objective:

- ☛ The subject helps the students to develop the fundamentals and basic concepts of Differential Equation. Students will be able to solve problems related to engineering applications by using these techniques.
- ☛ Apply the principles of Power series to solve a variety of practical problems in Engineering and Applied Science.
- ☛ Apply the concept of line, surface and volume integral with various theorems to solve practical problems.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: To introduce the basic concepts required to understand, construct, solve and interpret ordinary differential equations.
- ☛ CO 2: To teach methods to solve differential equations of various types.
- ☛ CO 3: To give an idea about Power series solutions; Legendre polynomials, Bessel functions.
- ☛ CO 4: To give an ability to apply knowledge of Partial Differential equation on engineering problems.
- ☛ CO 5: Formulate and solve problems related to vector calculus in the field of Industrial Organization Engineering.

Module 1: First order ordinary differential equations:

Exact, linear and Bernoulli's equations, Euler's equations, Equations not of first degree: equations solvable for p , equations solvable for y , equations solvable for x and Clairaut's type.

Module 2: Ordinary differential equations of higher orders:

Second order linear differential equations with variable coefficients, method of variation of parameters, Cauchy-Euler equation.

Module 3: Power series

Power series solutions; Legendre polynomials, Bessel functions of the first kind and their properties.

Module 4: Partial Differential Equations

Definition of Partial Differential Equations, First order partial differential equations, solutions of first order linear PDEs; Solution to homogeneous and non-homogeneous linear partial differential equations of second order by complementary function and particular integral method. Second-order

linear equations and their classification, Initial and boundary conditions, D'Alembert's solution of the wave equation; Heat diffusion and vibration problems, Separation of variables method to simple problems in Cartesian coordinates.

Module 5: Vector calculus

Gradient, curl and divergence ; Scalar line integrals, vector line integrals, scalar surface integrals, vector surface integrals, Theorems of Green, Gauss and Stokes.

Suggested Readings:

1. G.B. Thomas and R.L. Finney, *Calculus and Analytic geometry*, 9th Edition, Pearson, Reprint, 2002.
2. Erwin kreyszig, *Advanced Engineering Mathematics*, 9th Edition, John Wiley & Sons, 2006.
3. W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, 9th Edn., Wiley India, 2009.
4. S. L. Ross, *Differential Equations*, 3rd Ed., Wiley India, 1984.
5. E. A. Coddington, *An Introduction to Ordinary Differential Equations*, Prentice Hall India, 1995.
6. E. L. Ince, *Ordinary Differential Equations*, Dover Publications, 1958.
7. J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 7th Ed., Mc- Graw Hill, 2004.
8. N.P. Bali and Manish Goyal, *A text book of Engineering Mathematics*, Laxmi Publications, Reprint, 2008.
9. B.S. Grewal, *Higher Engineering Mathematics*, Khanna Publishers, 36th Edition, 2010.

Program: B.Tech

Semester: Second

Course: Programming for Problem Solving

Course Code: ESC103

Course Objective:

The objective of this course is to help the students

- ☛ To impart basic knowledge about simple algorithms, fundamentals of Programming Skills particularly in C Programming Language.
- ☛ To enable how to implement conditional branching, iteration and recursion.
- ☛ To enable them to use functions, arrays, pointers, strings and structures in solving problems.
- ☛ To understand how to solve problems related to matrices, searching, sorting, use files to perform read and write operations.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO1:** Understand the fundamentals of computer and C programming language, along with flow chart and algorithm to solve given program.
- ☛ **CO2:** Able to choose the loops and decision-making statements to solve the problem.
- ☛ **CO3:** Gain the knowledge of function, array , pointer and structure.
- ☛ **CO4:** Learn about implementation of file Operations in C programming for a given application.

Unit 1: Introduction to Programming (2 hrs)

Introduction to Programming (Flow chart/pseudocode, compilation etc.), Variables (including data types)

Unit 2: Arithmetic expressions and precedence (2 hrs)

Unit 3: Conditional Branching and Loops (8 hrs)

Writing and evaluation of conditionals and consequent branching, Iteration and loops

Unit 4: Arrays (6 hrs)

Arrays (1-D, 2-D), Character arrays and Strings

Unit 5: Basic Algorithms (6 hrs)

Searching, Basic Sorting Algorithms, Finding roots of equations, idea of time complexity

Unit 6: Function and Recursion (8 hrs)

Functions (including using built in libraries), Recursion with example programs such as Quick sort, Ackerman function etc.

Unit 7: Structure and Pointers (6 hrs)

Pointers, Structures (including self referential structures e.g., linked list, notional introduction)

Unit 8: File handling (2 hrs)

Suggested Reading:

1. E. Balagurusamy – *Programming in ANSI C*, 3rd Edn. , TMH, New Delhi, 2004
2. *Programming with C*, B.S.Gottfried (TMH)
3. Y. Kanetkar – *Let us C*, 4th Edition, BPB Publication, New Delhi; 2002
4. *The C Programming Language*, B.W. Kernighan, Dennis M.Ritchie, PHI/Pearson Education
5. *C Programming with problem solving*, J.A. Jones & K. Harrow, Dreamtech Press
6. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, Prentice

Program: B.Tech

Semester: Second

Course: Programming for Problem Solving

Course Code: ESC103P

Tutorial and Lab:

Tutorial 1: Problem solving using computers: Lab1: Familiarization with programming environment

Tutorial 2: Variable types and type conversions: Lab 2: Simple computational problems using arithmetic expressions

Tutorial 3: Branching and logical expressions: Lab 3: Problems involving if-then-else structures

Tutorial 4: Loops, while and for loops: Lab 4: Iterative problems e.g., sum of series

Tutorial 5: 1D Arrays: searching, sorting: Lab 5: 1D Array manipulation

Tutorial 6: 2D arrays and Strings, memory structure: Lab 6: Matrix problems, String operations

Tutorial 7: Functions, call by value: Lab 7: Simple functions

Tutorial 8 & 9: Numerical methods (Root finding, numerical differentiation, numerical integration): Lab 8 and 9: Numerical methods problems

Tutorial 10: Recursion, structure of recursive calls: Lab 10: Recursive functions

Tutorial 11: Pointers, structures and dynamic memory allocation Lab 11: Pointers and structures

Tutorial 12: File handling: Lab 12: File operations

Program: B.Tech

Semester: Second

Course: Workshop Practice

Course Code: ESC104

Course Objective:

- ☛ Students able to understand different tool & equipment for work shop practice.
- ☛ Students acquire skills for the preparation of different Carpentry/fitting/welding models.
- ☛ Students able to understand the safety precaution in the workshop
- ☛ Student acquires skills of Application orientated tasks.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Understand the concepts of various manufacturing methods such as casting, forming machining.
- ☛ CO 2: Understand the additive manufacturing with its implication in manufacturing.
- ☛ CO 3: Ability to Produce Fitting jobs as per specified dimensions with the use of various fitting tools.
- ☛ CO 4: Understand the basic concepts of electrical & electronics and understand the working principle of motors.
- ☛ CO 5: Ability to understand the recyclable plastics & non-recyclable plastics and various manufacturing process & also understand the various additives & fillers used in plastics & its implications.
- ☛ CO 6: Understand the various the types of welding such as Arc welding, TIG, MIG & various advanced welding & its practical implications in industrial arena.

Lectures & videos: (10 hours)

Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods (3 lectures)

CNC machining, Additive manufacturing (1 lecture)

Fitting operations & power tools (1 lecture)

Electrical & Electronics (1 lecture)

Carpentry (1 lecture)

Plastic moulding, glass cutting (1 lecture)

Metal casting (1 lecture)

Welding (arc welding & gas welding), brazing (1 lecture)

Suggested Reading:

1. *Workshop technology* by Hajara Chaudhary
2. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., “Elements of Workshop Technology”, Vol. I 2008 and Vol. II 2010, Media promoters and publishers private limited, Mumbai.
3. Kalpakjian S. And Steven S. Schmid, “Manufacturing Engineering and Technology”,
4. 4th edition, Pearson Education India Edition, 2002.
5. (iii) Gowri P. Hariharan and A. Suresh Babu, “Manufacturing Technology – I” Pearson Education, 2008.
6. Roy A. Lindberg, “Processes and Materials of Manufacture”, 4th edition, Prentice Hall India, 1998.
7. Rao P.N., “Manufacturing Technology”, Vol. I and Vol. II, Tata McGrawHill House, 2017.

Program: B.Tech

Semester: Second

Course: Workshop Practice Lab

Course Code: ESC104P

Workshop Practice: (60 hours)

- | | | |
|-------------------------------------|---|---|
| 1. Machine shop | - | 10 hours |
| 2. Fitting shop | - | 8 hours |
| 3. Carpentry | - | 6 hours |
| 4. Electrical & Electronics | - | 8 hours |
| 5. Welding shop | - | 8 hours (Arc welding 4 hrs + gas welding 4 hrs) |
| 6. Casting | - | 8 hours |
| 7. Smithy | - | 6 hours |
| 8. Plastic moulding & Glass Cutting | - | 6 hours |

Program: B.Tech

Semester: Second

Course: Environmental Science

Course Code: MC101

Course Objective:

- ☛ **Knowledge:** to help students, gain a variety of experiences and acquire a basic understanding of the environment and its associated problems.
- ☛ **Awareness:** to help students acquire an awareness of and sensitivity to the total environment and its allied problems.
- ☛ **Attitudes:** to help students to acquire a set of values and promote a feeling of concern for the environment and provide motivation for actively participating in environmental improvement and protection.
- ☛ **Participation:** to provide students with an opportunity to be actively involved at all levels, working towards the resolution of environmental problems.
- ☛ **Skills:** to help students to acquire the skills for identifying and solving environmental problems
- ☛ **Evaluation Ability:** to evaluate environmental measures and education programs in terms of ecological, economic, social and aesthetic factors.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Articulate the interconnected and interdisciplinary nature of environmental studies; Demonstrate an integrative approach to environmental issues with a focus on sustainability
- ☛ **CO 2:** Predict the consequences of human actions on the web of life, global economy and quality of human life, develop critical thinking for shaping strategies (scientific, social, economic and legal) for environmental protection and conservation of biodiversity, social equity and sustainable development
- ☛ **CO 3:** Acquire values and attitudes towards understanding complex environmental-economic social challenges, and participating actively in solving current environmental problems and preventing the future ones
- ☛ **CO 4:** Reflect critically on their roles, responsibilities, and identities as citizens, consumers and environmental actors in a complex, interconnected world, adopt sustainability as a practice in life, society and industry.

Unit I

Multidisciplinary nature of environmental studies, Natural Resources

Definition, scope and importance need for public awareness.

Renewable and non-renewable resources:

Natural resources and associated problems.

a) Forest resources: Use and over-exploitation, deforestation, case studies.

Timber extraction, mining, dams and their effects on forest and tribal people.

b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems

c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.

e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources, Case studies.

f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification.

- Role of an individual in conservation of natural resources.
- Equitable use of resources for sustainable lifestyles.

Unit II

Ecosystems, Biodiversity and its conservation

- Concept of an ecosystem.
- Structure and function of an ecosystem.
- Producers, consumers and decomposers.
- Energy flow in the ecosystem.
- Ecological succession.
- Food chains, food webs and ecological pyramids.
- Introduction, types, characteristic features, structure and function of the following ecosystem:-
 - a. Forest ecosystem
 - b. Grassland ecosystem
 - c. Desert ecosystem
 - d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)
- Introduction – Definition: genetic, species and ecosystem diversity.
- Bio geographical classification of India
- Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values
- Biodiversity at global, National and local levels.
- India as a mega-diversity nation
- Hot-spots of biodiversity.
- Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts.
- Endangered and endemic species of India
- Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

Unit III:

Environmental Pollution, Social Issues and the Environment

Definition, Cause, effects and control measures of:-

- a. Air pollution
 - b. Water pollution
 - c. Soil pollution
 - d. Marine pollution
 - e. Noise pollution
 - f. Thermal pollution
 - g. Nuclear hazards
- Solid waste Management: Causes, effects and control measures of urban and industrial wastes.
 - Role of an individual in prevention of pollution.
 - Pollution case studies.
 - Disaster management: floods, earthquake, cyclone and landslides.
 - From Unsustainable to Sustainable development
 - Urban problems related to energy
 - Water conservation, rain water harvesting, watershed management
 - Resettlement and rehabilitation of people; its problems and concerns. Case Studies
 - Environmental ethics: Issues and possible solutions.
 - Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust Case Studies.
 - Wasteland reclamation.
 - Consumerism and waste products.
 - Environment Protection Act.
 - Air (Prevention and Control of Pollution) Act.
 - Water (Prevention and control of Pollution) Act
 - Wildlife Protection Act
 - Forest Conservation Act
 - Issues involved in enforcement of environmental legislation.
 - Public awareness.

Unit IV

Human Population and the Environment, Field work

- Population growth, variation among nations.
- Population explosion – Family Welfare Programme.
- Environment and human health.
- Human Rights.
- Value Education.

- HIV/AIDS.
- Women and Child Welfare.
- Role of Information Technology in Environment and human health.
- Case Studies.
- Visit to a local area to document environmental assets-river/forest/grassland/hill/mountain
- Visit to a local polluted site-Urban/Rural/Industrial/Agricultural
- Study of common plants, insects, birds.
- Study of simple ecosystems-pond, river, hill slopes, etc.

Suggested Reading:

1. G. Kiely – Environmental Engineering Irwin/ McGraw Hill International Edition, 1997
2. M. L. Davis and S. J. Masen, Principles of Environmental Engineering and Science, McGraw Hill International Edition, 2004
3. E. Bharucha, Environmental Studies, University Grants Commission, New Delhi and Bharati Vidyapeeth Institute of Environment and Research, Pune

Program: B.Tech

Semester: Second

Course: Life Skills

Course Code: HSMC102

Course Objective:

The objectives of the course are:

- ☛ To familiarize students with basic aspect of Life Skills and its conceptual treatment, theoretical perspectives and practical strategies.
- ☛ To equip students with the social and interpersonal skills that enable them to cope with the demands of everyday life.
- ☛ To build self-confidence, encourage critical thinking, foster independence and help people to communicate more effectively.
- ☛ To enhance one's ability to meet the challenges of everyday life.
- ☛ To develop psychosocial competency.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: To develop Personal, Social and Professional Competency.
- ☛ CO 2: Gain Intellectual Competency.
- ☛ CO 3: Develop effective communication skills and practice effective written and oral communication
- ☛ CO 4: Develop wellness skills.
- ☛ CO 5: Develop life and career path.
- ☛ CO 6: Understand and develop professionalism and self responsibility.

Unit I: Know Thyself/ Discovering Self

- Introduction of the Subject, Learning Objective, Learning Outcome, Importance/ Benefit of the subject
- Know Thyself/ Discovering Self, Snap shot of yourself
- Assess your Personal Traits
- Strength and Weaknesses
- Hobbies and Interest
- Etiquettes and Manners {Personal Etiquette, Professional Etiquette, Social Etiquette , Tour & Travel}

Unit II: Grammar

- The Sentence, Kinds of Sentences
- Parts Of Speech
- Article

- Verb, Kinds Of Verb, Modal Auxiliaries'
- Tense
- Voice {Active and Passive}
- Narration (Direct and Indirect Speech)
- Common Errors

Unit III: Vocabulary

- Homophones,
- Word Formation {Prefix, Suffix}
- Word Power {Antonym, Synonym}
- Word Exercise
- Degree of Comparison
- Idioms and Phrases

Unit IV: Letter Writing

- Letter Writing, The Forms of Letter,
- Classification of Letters
- Social Letters
- Letter of Application
- Official Letters/Business Letters
- Editorial Letter

Unit V: Phonetics & Phonology BASICS

OF PHONETICS

Introduction

Speech Mechanism Phonetics

Symbols Classification of

Sounds

Consonants, Vowels & Diphthongs Stress

Pattern,

USAGE OF PHONETICS

Syllable, Syllable Division, Word Syllable Tone/

Accent/Pitch

Manner of Articulation Voice

Modulation

Suggested Books & Readings:

1. Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen, Leena. *Communication Skills ; Eastern Economy Edition*
5. Ghanekar, Dr. Anjali. *Essentials of Business Communication Skills ; Everest Publishing House*
6. David Green. *Contemporary English Grammar, Structure & Composition ; MacMillan*
7. *Dictionary; Oxford*
8. *Dictionary ; Longman*



RKDF UNIVERSITY RANCHI

MINING ENGINEERING

B.Tech



SYLLABUS
(2022-2026)
SEMESTER III



RKDF UNIVERSITY RANCHI

B.TECH SEMESTER III

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER III					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Basic Science Course	8BSC201	Mathematics III	100	4
2	Engineering Science Courses	8ESC201	Engineering Mechanics	100	3
3	Professional Core Courses	8PCCMiE201	Introduction to Mining	100	3
4	Professional Core Courses	8PCCMiE202	Mining Geology	100	3
5	Professional Core Courses	8PCCMiE203	Mine Surveying I	100	3
6	Humanities and Social Sciences	HSMC201	Effective Technical Communication	100	3
7	Mandatory Courses	MC201	Community Engagement and Social Responsibility	100	2
PRACTICAL/ SESSIONAL					
1	Professional Core Courses	8PCCMiE202P	Mining Geology Lab	50	1
2	Professional Core Courses	8PCCMiE203P	Mine Surveying I Lab	50	1
			TOTAL	800	23

Program: B.Tech

Semester: Three

Course: Mathematics III

Course Code: 8BSC201

Course Objective:

- ☛ The subject helps the students to develop the fundamentals and basic concepts in Laplace transform and solve problems related to engineering applications by using these techniques.
- ☛ To find the velocity and acceleration of a particle moving along a space curve.
- ☛ Apply the principles of counting, permutation, combination pigeon hole for solving practical problem.
- ☛ Use the knowledge of set theory, graph theory, function & relation in solving practical problem.
- ☛ Apply the concepts & theorems based on modern algebra (Group & Ring Theory).

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: - To introduce the basic concepts required to understand, construct, solve and interpret Laplace and Fourier Equation.
- ☛ CO 2: - Gain knowledge about set theory, relations, functions and their applications in real world.
- ☛ CO 3: - To give an ability to apply knowledge of discrete mathematics such as graph theory, permutation combination and important principles on engineering problems.
- ☛ CO 4: - Ability to use the concept of Group theory to solve practical problems of real world.

Module 1: Transform Calculus

Laplace Transform, Properties of Laplace Transform, Laplace transform of periodic functions. Finding inverse Laplace transform by different methods, convolution theorem. Evaluation of integrals by Laplace transform, solving ODEs by Laplace Transform method. Fourier transforms.

Module 2: Sets, Relations and Functions

Basic operations on sets, Cartesian products, disjoint union (sum), and power sets. Different types of relations, their compositions and inverses. Different types of functions, their compositions and inverses.

Module 3: Introduction to Counting Technique and graph :

Basic counting techniques –inclusion and exclusion, pigeon-hole principle, permutation, combination, summations. Introduction to recurrence relation and generating functions. Introduction to Graphs Graphs and their basic properties – degree, path, cycle, subgraph, isomorphism, Eulerian and Hamiltonian walk.

Module 4: Algebraic Structures

Algebraic structures with one binary operation – semi group, monoid and group. Cosets, Lagrange's theorem, normal subgroup, homomorphic subgroup. Congruence relation and quotient structures. Error correcting code. Algebraic structures with two binary operations ring, integral domain, and field. Boolean algebra and boolean ring (Definitions and simple examples only).

Suggested Reading:

1. *Higher Engineering Mathematics*, B.S. Grewa
2. N.P. Bali and Manish Goyal, *A text book of Engineering Mathematics*, Laxmi Publications, Reprint, 2010.
3. P. G. Hoel, S. C. Port and C. J. Stone, *Introduction to Probability Theory*, Universal Book Stall, 2003 (Reprint).
4. S. Ross, *A First Course in Probability*, 6th Ed., Pearson Education India, 2002.
5. J.P. Tremblay and R. Manohar, *Discrete Mathematical Structure and It's Application to Computer Science*", TMG Edition, TataMcgraw-Hill
6. Norman L. Biggs, *Discrete Mathematics*, 2nd Edition, Oxford University Press. *Schaum's Outlines Series*, Seymour Lipschutz, Marc Lipson.

Program: B.Tech

Semester: Three

Course: Engineering Mechanics

Course Code: 8ESC201

Course Objective:

- ☛ Use scalar and vector analytical techniques for analysing forces in statically determinate structures
- ☛ Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple, practical problems
- ☛ Apply basic knowledge of maths and physics to solve real-world problems
- ☛ Understand measurement error, and propagation of error in processed data
- ☛ Understand basic kinematics concepts – displacement, velocity and acceleration (and their angular counterparts);
- ☛ Understand basic dynamics concepts – force, momentum, work and energy;
- ☛ Understand and be able to apply Newton's laws of motion.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: To understand the basic concepts of force systems & its various force analysis.
- ☛ CO 2: Understand the static equilibrium of particles and rigid bodies both in two dimensions and three dimensions.
- ☛ CO 4: Analyze the properties of surfaces & solids in relation to moment of inertia.
- ☛ CO 5: Illustrate the laws of motion, kinematics of motion and their interrelationship.
- ☛ CO 6: Understand Kinetics of Rigid Bodies & Free and forced vibrations.

Module 1: *Introduction to Engineering Mechanics covering*, Force Systems Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy

Module 2: *Friction covering*, Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction, screw jack & differential screw jack;

Module 3: *Basic Structural Analysis covering*, Equilibrium in three dimensions; Method of Sections; Method of Joints; How to determine if a member is in tension or compression; Simple Trusses; Zero force members; Beams & types of beams; Frames & Machines;

Module 4: *Centroid and Centre of Gravity covering*, Centroid of simple figures from first principle, centroid of composite sections; Centre of Gravity and its implications; Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections; Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook.

Module 5: *Virtual Work and Energy Method*-Virtual displacements, principle of virtual work for particle and ideal system of rigid bodies, degrees of freedom. Active force diagram, systems with friction, mechanical efficiency. Conservative forces and potential energy (elastic and gravitational), energy equation for equilibrium. Applications of energy method for equilibrium. Stability of equilibrium.

Module 6: *Review of particle dynamics*-Rectilinear motion; Plane curvilinear motion(rectangular, path, and polar coordinates). 3-D curvilinear motion; Relative and constrained motion; Newton's 2nd law (rectangular, path, and polar coordinates). Work-kinetic energy, power, potential energy. Impulse-momentum (linear, angular); Impact (Direct and oblique).

Module 7:*Introduction to Kinetics of Rigid Bodies covering*, Basic terms, general principles in dynamics; Types of motion, Instantaneous centre of rotation in plane motion and simple problems; D'Alembert's principle and its applications in plane motion and connected bodies; Work energy principle and its application in plane motion of connected bodies; Kinetics of rigid body rotation;

Module 8:*Mechanical Vibrations covering*, Basic terminology, free and forced vibrations, resonance and its effects; Degree of freedom; Derivation for frequency and amplitude of free vibrations without damping and single degree of freedom system, simple problems, types of pendulum, use of simple, compound and torsion pendulums;

Tutorials *from the above modules covering*, To find the various forces and angles including resultants in various parts of wall crane, roof truss, pipes, etc.; To verify the line of polygon on various forces; To find coefficient of friction between various materials on inclined plane; Free body diagrams various systems including block-pulley; To verify the principle of moment in the disc apparatus; Helical block; To draw a load efficiency curve for a screw jack

Suggested Reading:

1. *A Textbook of Engineering Mechanics* By D S KUMAR
2. *A Textbook of Engineering Mechanics* By R S KHURMI
3. *A Textbook of Engineering Mechanics* By S.S. BHAVIKATTI
4. *A Textbook of Engineering Mechanics* By R.K.BANSAL

Program: B.Tech

Semester: Three

Course: Introduction to Mining

Course Code: 8PCCMiE201

Course Objective:

- ☛ Enable students to demonstrate the importance of mining in national economy.
- ☛ Students will understand the terminology associated with the discipline and be familiar with the available regulatory mechanism to enable safe & sustainable mining operations.
- ☛ Equips the students with detail knowledge on various engineering techniques used for drilling, blasting, roof support and allied activities in mine construction for exploitation of minerals.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Understand the history & significance of mining and mineral resources of India.
- ☛ CO 2: Gain the basic concept of mining, its terminology and classification.
- ☛ CO 3: Explain the technical knowledge of mode of access to deposit, type of mining & advantages & disadvantages.
- ☛ CO 4: Understand the different mining methods, its applicability & limitation.

Module 1

Significance of Mining Industry, History of Mining, Mining & Socio Economic impact. Mining and Surface Environment, Health and Safety. Mineral Resource of India, Mineral and energy resources of world, Terminology in Mining.

Module 2

Mining – definition and economic importance; mine – definition, different types and classification; Mine life cycle; Mineral deposit – different types and their classification.

Module 3

Modes of entry to a mine – shaft, incline, decline, adit and box-cut. Overview of surface mining: Types of surface mines, unit operations, basic bench geometry, applicability & limitations and advantages & disadvantages.

Module 4

Overview of underground mining: Different coal mining methods and their applicability & limitations; Different metal mining methods and their applicability & limitations; Basic concepts of transportation, ventilation, illumination and support in underground mines.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Explosives and Blasting Techniques*, S.K. Das
3. *Elements of mining technology*, D.J. Deshmukh

Program: B.Tech

Semester: Three

Course: Mining Geology

Course Code: 8PCCMiE202

Course Objective:

- ☛ This will provide basic knowledge on identification of various rocks & minerals by verifying the physical properties.
- ☛ The course will equip the students with the ability to appreciate the structure of earth and ore formations.
- ☛ The students will have knowledge about ore reserve estimation, ore assaying, remote sensing, geological mapping and identification of geological structures in the field.
- ☛ To practice the determination of engineering properties of rocks, preparation of weathering profiles, RMR, RQD, preparation of geological and structural maps and congestion of geological structures in the field.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Enhance the technical knowledge on shape, size, mass & density of earth, age of earth, structure of the earth.
- ☛ CO 2: Understand the concept of mineralogy with special light on sulphides, silicates and oxides minerals.
- ☛ CO 3: Explain the process of petrology and formation of rock.
- ☛ CO 4: Understand the concept of paleontology and stratigraphy. & coal and oil belts of India.

Module 1

Physical Geology: Origin of Earth, Interior of Earth, Study of Earth surface and features- Mountains/valley/rivers/lakes/volcanoes, study and their formation. Types of rocks and their process of formation, types of minerals.

Module 2

Mineralogy: Minerals: Physical and chemical properties of crystal, Types of crystallization system, classification of minerals and properties of common silicate minerals (Quartz Feldspar pyroxene Amphibole Garnet Olivine, Mica), Sulphides (Pyrite, Chalcopyrite Galena Sphalerite) and oxides (Hematite, Magnetite, Chromites, Pyrolusite Psilomelane).

Module 3

Petrology: Igneous rocks, Magma and lava extrusive and intrusive forms classification and description of some common igneous rock (Granite Dolerite Gabbros' Basalt Rhyolite Pegmatite) Sedimentary rocks, Sedimentation processes, classification and description of common Sedimentary rocks (Conglomerate Sandstone Shale Limestone). Metamorphic rock Processes of metamorphism, texture and structure of metamorphic rock, classification and description of some common metamorphic rock (Slate Pyrite Schist Gneiss, Quartzite, Marble)



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Module 4

Paleontology and Stratigraphy: Concept of paleontology, Fossils their mode of preservation and significance as indices of age and climate, concept of index fossils, Principle of stratigraphy, Broad stratigraphy subdivision and associated rock types of important coal belts and oil fields of India.

Suggested Reading:

1. *A text book of Geology, P.K. Mukherjee*
2. *Structural Geology, S.W. Chiplonkar*



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Program: B.Tech

Semester: Three

Course: Mine Geology Lab

Course Code: 8PCCMiE202P

List of Experiments:

1. To Study the geological maps
2. Mineralogy: Study of physical properties of Minerals.
3. Petrology: Study of common igneous rocks.
4. Petrology: Study of common sedimentary rocks.
5. Petrology : Study of common metamorphic rocks

Program: B.Tech

Semester: Three

Course: Mine Surveying I

Course Code: 8PCCMiE203

Course Objective:

- ☛ The course provides for basic skills in survey and correct usage of survey instruments.
- ☛ The student will be able to enter for the Elementary Survey Course.
- ☛ Appreciate the need for accurate and thorough note taking in field work to serve as a legal record
- ☛ Gain the ability to use modern survey equipment to measure angles and distances.
- ☛ Gain the ability to measure differences in elevation, draw and utilize contour plots, and calculate volumes for earthwork.
- ☛ Improve ability to function as a member of a survey party in completing the assigned field work.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Enhance the technical knowledge on linear measurements by chain surveying & tape surveying, compass surveying and plane table surveying.
- ☛ CO 2: Identify and evaluate the engineering problems in leveling and angular measurements with various surveying tools like theodolite, levels etc.
- ☛ CO 3: Apply the use of Total Stations & EDM in field and concept of contour.
- ☛ CO4: Understand the application of plane table survey and modern engineering tools necessary for mine surveying.

Module 1

Surveying: Definition, objective, classification and principles of surveying. Linear Measurement: Instruments for measuring distances; ranging and taping survey lines; Chain surveying-principle, field work, off-sets, booking and plotting, obstacles in taping.

Module 2

Angular Measurement: Bearing of lines; Rectangular coordinate system; Essentials of the micro-optic theodolite; Measurement of horizontal and vertical angles; Temporary and permanent adjustments; Theodolite traversing; Computation of co-ordinates; Adjustment of traverse; Temporary and permanent adjustments. Levelling: Definition terminology; Levelling instruments types- tilting, auto set and digital levels; Levelling staves; Different types of levelling-differential, profile, cross-sectional and reciprocal leveling; Booking and reduction methods; Underground leveling; Temporary and permanent adjustments of levels.

Module 3

Total Station: Principle of electronic measurement of distance and angles; construction and working with Total Station; Errors; Application and recent developments in Total Station. Contours: Concepts; Characteristics of contour; Contour Interval; Methods of contouring and uses of contours.

Module 4

Plane Table Surveying: Details surveying and contouring using plane table and micro-optic alidade.
Computation of area and volumes.

Suggested Reading:

1. *Surveying and leveling VOL I,II, T. P. Kanetkar*
2. *Surveying & Levelling , B.C. Punmia*

Program: B.Tech

Semester: Three

Course: Mine Surveying I Lab

Course Code: 8PCCMiE203P

List of Experiments:

1. Measurement of distance by Ranging and Chaining.
2. Locating various object by chain & cross staff survey.
3. Measurement of bearings of sides of traverse with prismatic compass and computation of correct included angle.
4. To find the difference in elevation between two points using Differential or Fly leveling.
5. Calculation of R.L. for different points involving 2 instrument stations & reduction by Height of Instrument & Rise and Fall methods.
6. Study of plane table surveying equipment's and accessories. Three point problem in plane table traversing.
7. Study of Auto Level and Dumpy Level; to study different parts of a Transit Theodolite and Temporary Adjustments.
8. To determine the distance between two inaccessible points by the help of a base length.

Program: B.Tech

Semester: Three

Course: Community Engagement and Social Responsibility

Course Code:

Course Objectives:

- ☛ To develop an appreciation of rural culture, life-style and wisdom amongst students.
- ☛ To learn about the status of various agricultural and rural development programmes.
- ☛ To understand causes for rural distress and poverty and explore solutions for the same.
- ☛ To apply classroom knowledge of courses to field realities and thereby improve quality of learning.

Course Outcomes:

After completing this course, student will be able to

- ☛ Gain an understanding of rural life, culture and social realities.
- ☛ Develop a sense of empathy and bonds of mutuality with local community.
- ☛ Appreciate significant contributions of local communities to Indian society and economy.
- ☛ Learn to value the local knowledge and wisdom of the community.
- ☛ Identify opportunities for contributing to community's socio-economic improvements.

Module 1:

Appreciation of Rural Society

(8 Hours)

Rural life style, rural society, caste and gender relations, rural values with respect to community, nature and resources, elaboration of "soul of India lies in villages" (Gandhi), rural infrastructure.

ASSIGNMENT: Prepare a map (physical, visual or digital) of the village you visited and write an essay about inter-family relations in that village.

Module 2:

Understanding rural economy & livelihood

(8 Hours)

Agriculture, farming, landownership, water management, animal husbandry, non-farm livelihoods and artisans, rural entrepreneurs, rural markets.

ASSIGNMENT: Describe your analysis of rural household economy, its challenges and possible pathways to address them.

Module 3:

Rural Institutions

(8 Hours)

Traditional rural organisations, Self-help Groups, Panchayati raj institutions (Gram Sabha, Gram Panchayat, Standing Committees), local civil society, local administration

ASSIGNMENT: How effectively are Panchayati raj institutions functioning in the village? What would you suggest to improve their effectiveness? Present a case study (written or audio- visual)

Module 4:

Rural Development Programmes

(8 Hours)

History of rural development in India, current national programmes: Sarva Shiksha Abhiyan, Beti Bachao, Beti Padhao, Ayushman Bharat, Swatchh Bharat, PM Awaas Yojana, Skill India, Gram Panchayat Decentralised Planning, NRLM, MNREGA, etc.

ASSIGNMENT: Describe the benefits received and challenges faced in the delivery of one of these programmes in the rural community; give suggestions about improving Implementation of the programme for the rural poor.

Suggested Reading:

1. Singh, Katar, *Rural Development : Principles, Policies and Management*, Sage Publications, New Delhi, 2015.
2. *A Hand book on Village Panchayat Administration*, Rajiv Gandhi Chair for Panchayati
3. *Raj Studies*, 2002.
4. United Nations, *Sustainable Development Goals*, 2015 un.org/sdgs/
5. M.P.Boraian, *Best Practices in Rural Development*, Shanlax Publishers, 2016.

Program: B.Tech

Semester: Three

Course: Effective Technical Communication

Course Code: HSMC201

Course Objective:

The objectives of the course are:

- ☛ To teach students the principles of technical communication for their academic and professional needs, focusing on essential written and oral skills for presenting technical information effectively.
- ☛ To make the students aware of the basic principles, which include the analysis of context, purpose and audience.
- ☛ To enhance fundamentals of technical report writing.
- ☛ To equip their effective technical presentations.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Be able to grasp the ideas and information put across in a communication situation.
- ☛ CO 2: To get a proper hold on communication techniques needed in a situation.
- ☛ CO 3: Have soft skills to work in a team for organizational development.
- ☛ CO 4: Be able to analyze information and interpret a case study for application.
- ☛ CO 5: To form simple reports and projects for organizational needs.

Module 1: Information Design and Development- Different kinds of technical documents, Information development life cycle, Organization structures, factors affecting information and document design, Strategies for organization, Information design and writing for print and for online media.

Module 2: Technical Writing, Grammar and Editing- Technical writing process, forms of discourse, Writing drafts and revising, Collaborative writing, creating indexes, technical writing style and language. Basics of grammar, study of advanced grammar, editing strategies to achieve appropriate technical style. Introduction to advanced technical communication, Usability, Human factors, Managing technical communication projects, time estimation, Single sourcing, Localization.

Module 3: Self Development and Assessment- Self assessment, Awareness, Perception and Attitudes, Values and belief, Personal goal setting, career planning, Self-esteem. Managing Time; Personal memory, Rapid reading, Taking notes; Complex problem solving; Creativity

Module 4: Communication and Technical Writing- Public speaking, Group discussion, Oral; presentation, Interviews, Graphic presentation, Presentation aids, Personality Development. Writing reports, project proposals, brochures, newsletters, technical articles, manuals, official notes, business letters, memos, progress reports, minutes of meetings, event report.

Module 5: Ethics- Business ethics, Etiquettes in social and office settings, Email etiquettes, Telephone Etiquettes, Engineering ethics, Managing time, Role and responsibility of engineer, Work culture in jobs, Personal memory, Rapid reading, Taking notes, Complex problem solving, Creativity.

Suggested Readings:

1. David F. Beer and David McMurrey, *Guide to writing as an Engineer*, John Willey. New York, 2004
2. Diane Hacker, *Pocket Style Manual*, Bedford Publication, New York, 2003. (ISBN 0312406843)
3. Shiv Khera, *You Can Win*, Macmillan Books, New York, 2003.
4. Raman Sharma, *Technical Communications*, Oxford Publication, London, 2004.
5. Dale Jungk, *Applied Writing for Technicians*, McGraw Hill, New York, 2004. (ISBN: 07828357-4)



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6. Sharma, R. and Mohan, K. *Business Correspondence and Report Writing*, TMH New Delhi 2002.

MINING ENGINEERING

B.Tech



SYLLABUS (2022-2026) SEMESTER IV

B.TECH SEMESTER IV

BTECH IN MINING ENGINEERING					
SEMESTER IV					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Engineering Science Courses	8ESC202	Mechanical Engineering	100	3
2	Professional Core Courses	8PCCMiE204	Drilling & Blasting	100	3
3	Professional Core Courses	8PCCMiE205	Geology for Mining Engineers	100	3
4	Professional Core Courses	8PCCMiE206	Mine Development	100	3
5	Professional Core Courses	8PCCMiE207	Mine Surveying II	100	3
6	Humanities and Social Sciences	HSMC202	Professional Practices, Laws & Ethics	100	3
7	Mandatory Courses	MC202	**Disaster Management	100	0
PRACTICAL/ SESSIONAL					
1	Professional Core Courses	8PCCMiE205P	Geology for Mining Engineers Lab	50	1
2	Professional Core Courses	8PCCMiE207P	Mine Surveying II Lab	50	1
			TOTAL	800	20
Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of IVth Semester Will be Credited in Vth Semester.					
** NOTE: Qualifying Non Credit Course					

Program: B.Tech

Semester: Four

Course: Mechanical Engineering

Course Code: 8ESC202

Course Objective:

- ☛ Ability to apply mathematics, science, and engineering
- ☛ Ability to design and conduct experiments, as well as to analyze and interpret data
- ☛ Ability to identify, formulate, and solve engineering problems
- ☛ Ability to apply modern engineering tools, techniques and resources to solve complex mechanical engineering activities with an understanding of the limitations.
- ☛ Ability to comprehend the thermodynamics and their corresponding processes that influence the behavior and response of structural components

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Understand the basic thermodynamics systems & concept of temperature and heat.
- ☛ CO 2: To identify and formulate power production based on the fundamentals laws of thermal engineering.
- ☛ CO 3: Understand the various cycles and its implications in real practical applications.
- ☛ CO 4: Understand the thermodynamics concepts of pure substances & properties of steam.
- ☛ CO 5: To appreciate concepts learnt in fundamentals laws of thermodynamics from which learning ideas how to sustain in energy crisis and think beyond curriculum in the field of alternative and renewable sources of energy. To communicate effectively the concepts of internal combustion engines and try to think beyond curriculum in alternative sources of energy.
- ☛ CO 6: Understand the concept of ideal & real gases & concept of gas mixtures & its practical applications in engineering.
- ☛ CO 7: Understand the Psychrometry and psychrometric charts, property calculations of air vapour mixtures & Psychrometric process..

Module 1:

Basic Concepts- Basic concepts - concept of continuum, macroscopic approach, Thermodynamic systems - closed, open and isolated. Property, state, path and process, quasistatic process, work, modes of work. Zeroth law of thermodynamics, concept of temperature and heat. Concept of ideal and real gases.

Module 2:

First Law of Thermodynamics- Concepts of Internal Energy, Specific HeatCapacities, Enthalpy. Energy Balance for Closed and Open Systems, Energy Balance for Steady-Flow Systems. Steady-Flow Engineering Devices. Energy Balance for Unsteady-Flow.

Module 3:

Second Law of Thermodynamics- Thermal energy reservoirs, heat engines energyconversion, Kelvin's and Clausius statements of second law, the Carnot cycle, the Carnot Theorem, the thermodynamic temperature scale, the Carnot heat engine, efficiency, the Carnot refrigerator and heat pump, COP. Clausius inequality, concept of entropy, principle of increase of entropy –

availability, the increase of entropy principle, perpetual-motion machines, reversible and irreversible processes, Entropy change of pure substances, isentropic processes, property diagrams involving entropy, entropy change of liquids and solids, the entropy change of ideal gases, reversible steady-flow work, minimizing the compressor work, isentropic efficiencies of steady-flow devices, and entropy balance. Energy a measure of work potential, including work potential of energy, reversible work and irreversibility, second-law efficiency, exergy change of a system, energy transfer by heat, work, and mass, the decrease of exergy principle and exergy destruction, energy balance: closed systems and control volumes energy balance.

Module 4:

Properties Of Pure Substance- Properties of pure substances. Thermodynamic properties of pure substances in solid, liquid and vapour phases. Phase rule, P-V, P-T, T-V, T-S, H-S diagrams, PVT surfaces. Thermodynamic properties of steam. Calculations of work done and heat transfer in non-flow and flow processes.

Module 5:

Power Cycles- Vapour and combined power cycles, including the Carnot vapor cycle, Rankine cycle: the ideal cycle for vapor power, the ideal reheat and regenerative and the second-law analysis of vapour power cycles. Gas power cycles, including basic considerations in the analysis of power cycles, the Carnot cycle and its value in engineering, an overview of reciprocating engines, air standard assumptions, gasoline engine Otto cycle, diesel engine cycle, gas-turbine Brayton cycle, and the second-law analysis of gas power cycles.

Module 6:

Ideal and Real Gases and Thermodynamic Relations- Gas mixtures – properties ideal and real gases. Equation of state, Avogadro's Law, Vander Waal's equation of state, Compressibility factor, compressibility chart. Dalton's law of partial pressure. Exact differentials, T-D relations, Maxwell's relations. Clausius Clapeyron equations, Joule – Thomson coefficient.

Module 7:

Psychrometry and psychrometric charts, property calculations of air vapour mixtures. Psychrometric process – Sensible heat exchange processes. Latent heat exchange processes. Adiabatic mixing, evaporative cooling. Use of standard thermodynamic tables, Mollier diagram, Psychrometric chart and Refrigerant property tables. Refrigeration cycles, including refrigerators and heat pumps, the ideal reversed Carnot vapour-compression refrigeration cycle, actual vapor-compression refrigeration cycles, heat pump systems, gas refrigeration cycles, and absorption refrigeration systems.

Suggested Reading:

1. Nag.P.K., "Engineering Thermodynamics", Tata McGraw-Hill, New Delhi.
2. Cengel, Thermodynamics – An Engineering Approach Tata McGraw Hill, New Delhi.
3. Sonntag, R. E., Borgnakke, C., & Wylen, G. J. V. Fundamentals of thermodynamics: Wiley.
4. Moran, M. J., Shapiro, H. N., Boettner, D. D., & Bailey, M. Fundamentals of
5. Jones, J. B., & Dugan, R. E. Engineering thermodynamics: Prentice Hall.

Program: B.Tech

Semester: Four

Course: Drilling and Blasting

Course Code: 8PCCMiE204

Course Objective:

The objective of this course is to help the students

- ☛ To develop knowledge on drilling and blasting operations in underground and surface mines.
- ☛ To design blasting pattern for mines and blast design in surface and underground mines.
- ☛ To develop knowledge about exploration techniques, drilling operations, casing, various explosives used in opencast & underground.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Understand the exploration techniques, drilling operations, casing, coring and data interpretation.
- ☛ **CO 2:** Analyze the appropriate application of explosive in opencast & underground mines.
- ☛ **CO 3:** Explain blasting engineering for understanding, formulating and solving blast hole design problems in surface mining.
- ☛ **CO 4:** Design various drilling & blasting techniques and their applicability in underground mines.

Module 1

(8 Hours)

Exploratory Drilling : Drilling for exploration; Various types of exploratory drills and their applicability- Auger, Cable-tool, Odex, Core Drills; Core recovery; single and double tube core barrels, wire line core barrel; Storage of cores; Interpretation of borehole data.

Module 2

(8 Hours)

Explosives and Initiation systems : Types of explosives, their composition and properties, classification; Selection of explosives; Manufacture, transport, storage and handling of explosive; Testing of explosives; Types of initiating systems- Electrical Detonators, Detonator fuse, Detonator Relays, NONEL, Electronic Detonators, Blasting accessories, exploders.

Module 3

(10 Hours)

Drilling & Surface Mines: Drilling, Blast hole drills –types, classification, applicability and limitations; Mechanics of drilling, performance parameters, drilling cost, compressed air requirement for hole cleaning; Selection of drilling systems, drilling errors, organization of drilling. Blasting: Mechanics of rock fragmentation; Living stone theory of crater formation; Factors affecting blasting, Blast design –estimation of burden and spacing, estimation of charge requirement; Initiation patterns; Secondary blasting – pop and plaster shooting; Problems associated with blasting, Ground vibration and air over pressure, Blast instrumentation.

Module 4

(10 Hours)

Drilling & Blasting in Underground Mines: Coalmines: Drilling systems and their applicability, blasting-off-solid, different blasting cuts, ring hole blasting, calculation of specific charge, specific drilling and detonator factor, initiation patterns. Metal mines: Drilling systems and their applicability, blast design for horizontal drivages, different blasting cuts, long hole blasting, and vertical crater retreat blasting.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Explosives and Blasting Techniques*, S.K. Das
3. *Elements of Mining Technology Vol 1*, D J Deshmukh
4. *Underground Practices*, R D Singh

Program: B.Tech

Semester: Four

Course: Geology for Mining Engineers

Course Code: 8PCCMiE205

Course Objective:

The objective of this course is to help the students

- ☛ To develop knowledge on economics of ore, exploration and practical site investigation.
- ☛ To create knowledge about geology of rock, geological disturbances and identification of geological structures in the field.
- ☛ To know the coal geology and complete geology of petroleum from origin to extraction.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Discuss the structural geology in aspects of engineering requirements & structural disturbances.
- ☛ **CO 2:** Apply the knowledge gained in the contexts of exploration and mineral distribution and able to design the stages of exploration of economic minerals.
- ☛ **CO 3:** Discuss the geology of coal along with its classification and origin of coal.
- ☛ **CO 4:** Describe the concept on petroleum geology with its migration, accumulation & important basins of India.

Module I

(10 Hours)

Structural Geology: Study of topographical maps; Altitude of planar and linear structures; Effects of topography on outcrops. Unconformities, folds, faults and joints - their nomenclature, classification and recognition. Forms of igneous intrusions - dyke, sill and Batholith. Effects of folds and fractures on strata/ore bodies and their importance in mining operations. Principles of stereographic projections of linear and planar features of rocks.

Module 2

(10 Hours)

Economic Geology and Exploration Geology: Introduction and scope of economic geology; Ore and gangue; Processes of ore formation; Major Indian mineral deposits (Iron, Manganese, Copper, Lead, Zinc) - distribution and mode of occurrence. Mineral Exploration – concepts and methods viz. surface and subsurface; Exploration strategy and design; Stages of exploration; Resources and reserves.

Module 3

(8 Hours)

Coal Geology : Rank, characteristics and important constituents of coal; Classification and origin of coal; Chief characteristics of Indian coals; Geology of the principal coalfields of India.

Module 4

(8 Hours)

Petroleum Geology : Concept of organic constituents of petroleum origin, migration, accumulation, concept of traps and important petroliferous basins of India.

Suggested Reading:



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1. *A text book of Geology, P.K. Mukherjee*
2. *Physical Geology, A.K. Dutta*

Program: B.Tech

Semester: Four

Course: Geology for Mining Engineers Lab

Course Code: 8PCCMiE204P

Course Objective:

1. Study of topographic maps.
2. Completion of outcrops: 1– point and 3–point problems; Map illustrating ‘V’ rules.
3. Calculation of attitude, thickness and depth of ore bodies.
4. Fracture patterns in rose diagram.
5. Maps illustrating fold, fault and unconformity.
6. Stereographic projection.

Program: B.Tech

Semester: Four

Course: Mine Development

Course Code: 8PCCMiE206

Course Objective:

The objective of this course is to help the students

- ☛ To develop basic insight to access of deposit methods & shaft sinking along with drifting and tunneling technology.
- ☛ To incorporate the understanding of methods of driving underground openings & headings.
- ☛ To appraise the special mining techniques and problems related to sinking & tunneling.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Understand the various methods of development, opening with their shape & locations of coal and metal mining.
- ☛ **CO 2:** Explain the various methods, equipments and operations of shaft sinking
- ☛ **CO 3:** Gain the knowledge of drivage of horizontal openings in underground and tunneling.
- ☛ **CO 4:** Learn about the recent developments in shaft sinking & surface plants in underground mines.

Module 1

(12 Hours)

Opening-up of Deposits: Choice of mode of entry, shaft, decline and combined mode, their applicability, number and disposition. Vertical and Inclined Shafts: Location, shape, size, and organization of shaft sinking, construction of shaft collar, shaft fittings. Shaft Sinking Operations: Ground breaking and muck disposal, tools and equipment, lining; ventilation, lighting and watering; sinking in difficult and water-bearing ground.

Module 2

(8 Hours)

Insets: Design, excavation and lining. Mechanized Sinking: Simultaneous sinking and lining; slip-form method of lining; high speed sinking, Shaft Boring: Methods and equipment.

Module 3

(10 Hours)

Special Attributes: Widening and deepening of inclined and vertical shafts; staple shafts, raised shafts. Main Haulage Drifts and Tunnels: Purpose, shape, size and location; excavation ground breaking, muck disposal, ventilation and supporting. High Speed Drifting/Tunneling: Application of mechanized methods; road headers and tunnel boring machines.

Module 4

(6 Hours)

Recent developments in shaft sinking and drifting/tunneling. Layouts of pit-top and pit-bottom, Coal Handling Plant, Bunkers and Railway Sidings.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Explosives and Blasting Techniques*, S.K. Das
3. *Introductory mining engg* by H.L.Hartman

Program: B.Tech

Semester: Four

Course: Mine Surveying II

Course Code: 8PCCMiE207

Course Objective:

The objective of this course is to help the students

- ☛ To understand & carry out underground surveys and stope measuring.
- ☛ To develop plotting contour plans
- ☛ Understand advanced surveying techniques such as remote sensing and geodetic surveying.
- ☛ The students will have practical knowledge about recent development of surveying and mine planning.
- ☛ Students are made aware about the modern survey equipment and methods for precision survey.
- ☛ The students will have practical knowledge about survey instruments and its workings.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ **CO 1:** Illustrate & apply various survey instruments like Total Station, Theodolite, EDM used in general and mine survey, through practical and demonstrations.
- ☛ **CO 2:** Apply knowledge of surveying for understanding, formulating and solving correlation surveying problems & slope surveying.
- ☛ **CO 3:** Demonstrate GPS used in general mine survey in opencast mine.
- ☛ **CO 4:** Analyze the application of IT in surveying & GIS.

Module 1

(10 Hours)

Control Surveys: Triangulation–classification; Reconnaissance; Procedures for angles and Base-Line measurement; Comparison with precise EDM traversing. Direction Measurement by Astronomical observation: Definition of astronomical terms; Time systems; Determination of true bearing of a survey line by astronomical observations. Gyro-North Determination: Principle of Gyro-theodolite/ Gyromat; Determination of Gyro-north.

Module 2

(12 Hours)

Correlation: Methods of correlation–direct traversing in inclined shaft, correlation in vertical shaft–single and two shafts, Gyro-Laser combination; Shaft depth measurement. Development Surveys: Control of direction and gradient in drifts, tunnels, raises, winzes. Slope Surveying: Purpose; Methods of survey in moderately and steeply inclined ore bodies, flat and vertical ore bodies/seams.

Module 3

(8 Hours)

Slope Monitoring in Opencast Mines: Geodetic and Remote Sensing Methods, Slope Stability Radars GPS: Principle of GPS; Instrument; Errors and working with GPS; Application of GPS in mine surveying; Developments in satellite based Navigation system.

Module 4

(6 Hours)



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Application of Automation& IT in surveying: Data acquisitions; Preparation of plans and sections; Calculation of earth works. Introduction to surveying software's. Application of GIS and Remote Sensing in Surveying.

Suggested Reading:

1. *Surveying and leveling , T. P. Kanetkar*
2. *Surveying & Levelling , B.C. Punmia*



Program: B.Tech

Semester: Four

Course: Mine Surveying II Lab

Course Code: 8PCCMiE207P

List of Experiments:

1. Precise Levelling and distribution of closing error.
2. Study and sketch of Digital Level.
3. Detailed surveying and contouring with Total Station.
4. Working with SURPAC and preparation of Mine Plan.
5. Earthwork calculation using SURPAC and plotting.
6. Plotting of sections using LISCAD.
7. Study and practice of Gyro-theodolite/Gyromat.
8. Study of GPS and data collection; GPS data downloading and post- processing.
9. Study of Nadir Plummet and LASER for Correlation survey.
10. Subsidence monitoring using 1" – Micro-optic Theodolite & Precise Level.

Program: B.Tech

Semester: Four

Course: Disaster Management

Course Code: MC201

Course Objective:

- ☛ Develop an understanding of the key concepts, definitions a key perspectives of All Hazards Emergency Management
- ☛ Understand the Emergency/Disaster Management Cycle
- ☛ Have a basic understanding for the history of Emergency Management
- ☛ Develop a basic under understanding of Prevention, Mitigation, Preparedness, Response and Recovery
- ☛ Develop a basic understanding for the role of public a private partnership.

Course Outcome:

- ☛ Understanding foundations of hazards, disasters and associated natural/social phenomena
- ☛ Familiarity with disaster management theory (cycle, phases)
- ☛ Knowledge about existing global frameworks and existing agreements (e.g. Sendai)
- ☛ Methods of community involvement as an essential part of successful DRR
- ☛ Humanitarian Assistance before and after disaster
- ☛ Technological innovations in Disaster Risk Reduction: Advantages and problems
- ☛ Experience on conducting independent DM study including data search, analysis and presentation of disaster case study.
- ☛ Respond to disaster risk reduction initiatives and disasters in an effective, humane and sustainable manner.

Module I

Understanding Disasters:

Understanding the concepts and definitions of disaster, hazard, vulnerability, risk, importance, dimensions & scope of Disaster Management, Disaster Management cycle and disaster profile of India.

Module II

Types, Trends, Causes, Consequences and Control of Disaster: Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunderstorms, hail storms, avalanches, droughts, cold and heat waves); Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear, bomb threat, explosion) and Man-made Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters; terrorist attack, , sudden shooting); Global Disaster Trends–Emerging Risks of Disasters–Climate Change and Urban Disasters; Financial emergency(risk of eviction, risk in arrears, sudden health emergency, family emergency, unexpected loss of income).

Module III

Prevention and Mitigation of Disaster :Disaster Mitigation: meaning and concept, Disaster Mitigation Strategies Emerging Trends in Disaster Mitigation, Mitigation management, Role of Team and Coordination. Disaster Preparedness: Concept & Nature, Disaster Preparedness Plan, Preventions. Roles & Responsibilities of Different Agencies and Government, Technologies for

Disaster Management. Early Warning System; Preparedness, Capacity Development; Awareness during Disaster.

Module IV

Applications of Science and Technology for Disaster Management & Mitigation: Geo-informatics in Disaster Management (RS, GIS, GPS and RS) Disaster Communication System (Early Warning and Its Dissemination) Land Use Planning and Development, Regulations, Disaster Safe Designs and Constructions, Structural and Non Structural Mitigation of Disasters.

Suggested Reading:

1. *Disaster Management- J. P. Singhal, Laxmi Publications.*
2. *Disaster Management - Dr. Mrinalini Pandey, Wiley India Pvt. Ltd.*
3. *Disaster Science and Management- Tushar Bhattacharya, McGraw Hill Education (India) Pvt. Ltd.*
4. *Disaster Management: Future Challenges and Opportunities - Jagbir Singh, K W Publishers Pvt. Ltd.*

Program: B.Tech

Semester: Four

Course: Professional Practices, Laws & Ethics

Course Code: HSMC202

Course Objective:

To impart basic skills of Professional Communication in English through basic elements of civil engineering professional practice are introduced in this course. Roles of all participants in the process-owners, developers, designers, consultants, architects, contractors, and suppliers - are described. Basic concepts in professional practice, business management, public policy, leadership, and professional licensure are introduced. The course covers professional relations, civic responsibilities, and ethical obligations for engineering practice. The course also describes contracts management, and various legal aspects related to engineering. Further, the course familiarizes students with elementary knowledge of laws that would be of utility in their profession, including several new areas of law such as IPR, ADR.

Course Outcome:

- ☛ Know the definitions and major concepts in ethics
- ☛ Understand the ethical approach to decision making
- ☛ Follow the assessment criteria steps when making a decision.
- ☛ Understand the Code of Conduct.
- ☛ Explain the concept of personal ethics and their relationship to business and engineering ethics.
- ☛ Describe the role of a code of conduct in a work environment, when following ethical principles.
- ☛ Uphold the code of conduct within the work team.
- ☛ Identify and solve ethical problems in upholding the code of conduct
- ☛ Communicate effectively, when explaining and describing the concept of ethics and the application of a code of conduct.

Module 1 A: Professional Practice – Respective roles of various stakeholders: Government (constituting regulatory bodies and standardization organizations, prescribing norms to ensure safety of the citizens); Standardization Bodies (ex. BIS, IRC)(formulating standards of practice); professional bodies (ex. Institution of Engineers(India), Indian Roads Congress, IIA/ COA, ECI, Local Bodies/ Planning Authorities) (certifying professionals and offering platforms for interaction); Clients/ owners (role governed by contracts); Developers (role governed by regulations such as RERA); Consultants (role governed by bodies such as CEAI); Contractors (role governed by contracts and regulatory Acts and Standards); Manufacturers/ Vendors/ Service agencies (role governed by contracts and regulatory Acts and Standards)

Module 1 B: Professional Ethics – Definition of Ethics, Professional Ethics, Business Ethics, Corporate Ethics, Engineering Ethics, Personal Ethics; Code of Ethics as defined in the website of Institution of Engineers (India); Profession, Professionalism, Professional Responsibility,

Professional Ethics; Conflict of Interest, Gift Vs Bribery, Environmental breaches, Negligence, Deficiencies in state-of-the-art; Vigil Mechanism, Whistle blowing, protected disclosures.

Module 2: General Principles of Contracts Management: Indian Contract Act, 1972 and amendments covering General principles of contracting; Contract Formation & Law; Privacy of contract; Various types of contract and their features; Valid & Voidable Contracts; Prime and sub-contracts; Joint Ventures & Consortium; Complex contract terminology; Tenders, Request For Proposals, Bids & Proposals; Bid Evaluation; Contract Conditions & Specifications; Critical /“Red Flag” conditions; Contract award & Notice To Proceed; Variations & Changes in Contracts; Differing site conditions; Cost escalation; Delays, Suspensions & Terminations; Time extensions & Force Majeure; Delay Analysis; Liquidated damages & Penalties; Insurance & Taxation; Performance and Excusable Non-performance; Contract documentation; Contract Notices; Wrong practices in contracting (Bid shopping, Bid fixing, Cartels); Reverse auction; Case Studies; Build-Own-Operate & variations; Public-Private Partnerships; International Commercial Terms;

Module 3 : Arbitration, Conciliation and ADR (Alternative Dispute Resolution) system: Arbitration – meaning, scope and types – distinction between laws of 1940 and 1996; UNCITRAL model law – Arbitration and expert determination; Extent of judicial intervention; International commercial arbitration; Arbitration agreements – essential and kinds, validity, reference and interim measures by court; Arbitration tribunal – appointment, challenge, jurisdiction of arbitral tribunal, powers, grounds of challenge, procedure and court assistance; Award including Form and content, Grounds for setting aside an award, Enforcement, Appeal and Revision; Enforcement of foreign awards – New York and Geneva Convention Awards; Distinction between conciliation, negotiation, mediation and arbitration, confidentiality, resort to judicial proceedings, costs; Dispute Resolution Boards; Lok Adalats

Module 4 : *Engagement of Labour and Labour & other construction-related Laws:* Role of Labour in Civil Engineering; Methods of engaging labour- on rolls, labour sub-contract, piece rate work; Industrial Disputes Act, 1947; Collective bargaining; Industrial Employment (Standing Orders) Act, 1946; Workmen’s Compensation Act, 1923; Building & Other Construction Workers (regulation of employment and conditions of service) Act (1996) and Rules (1998); RERA Act 2017, NBC 2017

Module 5 : *Law relating to Intellectual property:* Introduction – meaning of intellectual property, main forms of IP, Copyright, Trademarks, Patents and Designs, Secrets; Law relating to Copyright in India including Historical evolution of Copy Rights Act, 1957, Meaning of copyright – computer programs, Ownership of copyrights and assignment, Criteria of infringement, Piracy in Internet – Remedies and procedures in India; Law relating to Patents under Patents Act, 1970 including Concept and historical perspective of patents law in India, Patentable inventions with special reference to biotechnology products, Patent protection for computer programs, Process of obtaining patent – application, examination, opposition and sealing of patents, Patent cooperation treaty and

grounds for opposition, Rights and obligations of patentee, Duration of patents – law and policy considerations, Infringement and related remedies;

Suggested Readings:

1. B.S. Patil, *Legal Aspects of Building and Engineering Contracts*, 1974.
2. *The National Building Code*, BIS, 2017
3. *RERA Act*, 2017
4. Meena Rao (2006), *Fundamental concepts in Law of Contract*, 3rd Edn. Professional Offset
5. Neelima Chandiramani (2000), *The Law of Contract: An Outline*, 2nd Edn. Avinash Publications Mumbai
6. Avtarsingh (2002), *Law of Contract*, Eastern Book Co.
7. Dutt (1994), *Indian Contract Act*, Eastern Law House
8. Anson W.R. (1979), *Law of Contract*, Oxford University Press
9. Kwatra G.K. (2005), *The Arbitration & Conciliation of Law in India with case law on UNCITRAL Model Law on Arbitration*, Indian Council of Arbitration
10. Wadhera (2004), *Intellectual Property Rights*, Universal Law Publishing Co.
11. T. Ramappa (2010), *Intellectual Property Rights Law in India*, Asia Law House
12. Bare text (2005), *Right to Information Act*
13. O.P. Malhotra, *Law of Industrial Disputes*, N.M. Tripathi Publishers
14. K.M. Desai (1946), *The Industrial Employment (Standing Orders) Act*
15. Rustamji R.F., *Introduction to the Law of Industrial Disputes*, Asia Publishing House
16. Vee, Charles & Skitmore, Martin (2003) *Professional Ethics in the Construction Industry*, Engineering Construction and Architectural management, Vol.10, Iss2, pp 117-127, MCB UP Ltd
17. American Society of Civil Engineers (2011) *ASCE Code of Ethics – Principles Study and Application*
18. *Ethics in Engineering*- M.W.Martin & R.Schinzinger, McGraw-Hill
19. *Engineering Ethics*, National Institute for Engineering Ethics, USA
20. www.ieindia.org
21. *Engineering ethics: concepts and cases* – C. E. Harris, M.S. Pritchard, M.J.Rabins
21. *CONSTRUCTION CONTRACTS*, <http://www.jnormanstark.com/contract.htm>
22. *Internet and Business Handbook*, Chap 4, *CONTRACTS LAW*
23. <http://www.laderapress.com/laderapress/contractslaw1.html>
24. *Contract & Agreements*
25. <http://www.tco.ac.ir/law/English/agreements/General/Contract%20Law/C.htm>
26. *Contracts*, <http://206.127.69.152/jgretch/crj/211/ch7.ppt>
27. *Business & Personal Law*. Chapter 7. “How Contracts Arise”, <http://yucaipahigh.com/schristensen/lawweb/lawch7.ppt>
28. *Types of Contracts*, <http://cmsu2.cmsu.edu/public/classes/rahm/meiners.con.ppt>
29. TYPES OF CONTRACTS AND IMPORTANT PROVISIONS, <http://www.worldbank.org/html/opr/consult/guidetxt/types.html>
30. *Contract Types/Pricing Arrangements Guideline- 1.4.G (11/04/02)*, <http://www.sandia.gov/policy/14g.pdf>

MINING ENGINEERING

B.Tech



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SYLLABUS (2022-2026) SEMESTER V

B.TECH SEMESTER V

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER V					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Professional Core Courses	8PCCMiE301	Mining Machinery I	100	3
2	Professional Core Courses	8PCCMiE302	Surface Mining	100	3
3	Professional Core Courses	8PCCMiE303	Underground Coal Mining	100	3
4	Professional Core Courses	8PCCMiE304	Rock Mechanics	100	3
5	Professional Core Courses	8PCCMiE305	Numerical & Statistical Methods	100	3
6	Professional Core Courses	8PCCMiE306	Remote Sensing & GIS	100	3
7	Humanities and Social Sciences	HSMC301	**Professional Skills	100	0
8	Mandatory Courses	MC301	**Constitution of India	100	0
PRACTICAL / SESSIONAL					
1	Professional Core Courses	8PCCMiE304P	Rock Mechanics Lab	50	1
2	Project	8PROJMiE301	Vocational Training Report I	100	2
			TOTAL	950	21
Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of Vth Semester Will be Credited in VII th Semester.					
**NOTE: Qualifying Non Credit Course.					

Program: B.Tech

Semester: Five

Course: Mining Machinery I

Course Code: 8PCCMiE301

Course Objective:

- ☛ Enables the students to select appropriate machinery for various mining operations based on the production targets.
- ☛ The students will have knowledge on function of winding engines, winding accessories, pit-top and bottom mine circuits.
- ☛ They will also know about working of various coal face machinery, face haulage systems and conveyors.
- ☛ The students will have basic knowledge on motive power used in mines, rope haulage and other transport systems.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the classification & application of mine machinery.
- ☛ CO2: Understand the winding system and safety features.
- ☛ CO3: Able to apply the knowledge of power transmission system and installation in mines.
- ☛ CO4: Explain the types, selection and uses of mining ropes and able to apply the knowledge of rope capping and splicing.

Module 1

Classification, application, constructional features of drilling machines used in coal and metal mining, Coal cutters, shearer, plough, continuous miner, road header and dirt header. Loading and transport equipment, man riding systems.

Module 2

Classification and constructional difference of different types of winders, mechanics of winding, power calculation, rope selection, inspection and maintenance. Safety features and contrivances. Classification, construction and selection steps for mine dewatering pumps.

Module 3

Transmission of Power Belt, rope, chain, gear, hydraulic and electro-hydraulic transmission. Compressed Air Comparison with other sources of power. Air compressors – types, construction, installation and maintenance. Compressed air transmission and distribution, compressed air drills, pneumatic picks, air motors and other compressed air equipment.

Module 4

Wire Ropes Types, construction and uses. Rope deterioration and maintenance. Capping and splicing of rope. Rope haulages. Track, mine tubs and cars. Safety appliances on haulage roads. Locomotive haulage. Mono rail. Statutory Provisions

Suggested Reading:

1. D.J. Deshmukh VOL III
2. Mine pump, haulage, winding. S. GHATAK

Program: B.Tech

Semester: Five

Course: Surface Mining

Course Code: 8PCCMiE302

Course Objective:

- ☛ Enables the students to apply the knowledge of mining machinery in their economic deployment for achieving the production targets in opencast mines.
- ☛ The students will have insight about the advanced techniques for mine planning, geotechnical investigation and equipment management and also will understand the modern trends in opencast mines, safety and environment.
- ☛ The students will have ability to classify and select the suitable surface mining methods and equipment based on site conditions. They will also have a concept of waste dump formations and slope failures in surface mines.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the basic concept of surface mining for understanding the applicability & slope stability problem in any opencast mine.
- ☛ CO2: Understand the application and operation of shovel, dragline, dozer and other opencast machinery.
- ☛ CO3: Understand the application, operation and limitation of bucket wheel excavator and surface transport system.
- ☛ CO4: Explain the applicability of the different in-pit crushing and conveying methods & interpretation of slope stability.

Module 1

Introduction: Surface mining-basic concepts, applicability, advantages and disadvantages; Role of surface mining in total mineral production; Surface mining unit operations; Surface mining equipment systems— classification, applicability, advantages and disadvantages. Opening up of deposits: Boxcut—objective, types, parameters, methods; Factors affecting selection of box cut site; Production benches— formation, parameters and factors affecting their selection.

Module 2

Discontinuous/cyclic methods of excavation and transport: Shovel: Applicability and limitations of shovel, Cycle time and productivity calculation for shovel; Dragline operation: Applicability and limitations, different modes of operation; Cycle time and productivity calculation; Maximum usefulness factor and its significance in selection of dragline for a given situation; Scrapers: Applicability and limitations, various types; Method and cycle of operation; Dozers: Applicability and limitations; Types and classification; merits and demerits; Method and cycle of operation. Front-end- loaders: Applicability and limitations.

Module 3

Continuous methods of excavation and transport: Bucket wheel excavators: Applicability and limitations; Types and principle of operation; Calculation of productivity. Continuous surface miners: Types, classification, applicability and limitations; Principles of operation; Conveyors: types of conveyors; Mode of operation, applicability and limitations; Merits and demerits of conveyor as a system of transportation.

Module 4



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Different in-pit crushing and conveying methods and their respective applicability & limitations, Types of mine slope—high wall and waste dumps; Common modes of slope failure; Factors influencing stability of slopes; Slope stability assessment techniques.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Explosives and Blasting Techniques*, S.K. Das
3. *Surface Mining*, G.B. Mishra

Program: B.Tech

Semester: Five

Course: Underground Coal Mining

Course Code: 8PCCMiE303

Course Objective:

- ☛ The students will have good exposure about the various advanced methods of coal mining with the knowledge about advanced coal face mechanization.
- ☛ To pioneer the history of longwall mining and its development stages.
- ☛ To understand the extraction, support and transport on a longwall face.
- ☛ Enables the students to appreciate various coal mining methods and to ably manage highly mechanized mines.
- ☛ To introduce the recent trends of level of mechanization for coal face.
- ☛ To understand the various advanced methods of coal mining.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Explain the technical knowledge on development & extraction of coal by board & pillar mining.
- ☛ CO2: Understand the method & problems of extraction of coal from the underground mines along with stowing.
- ☛ CO3: Explain the basic knowledge of long wall mining with its applicability, components and machinery required for mining.
- ☛ CO4: Apply the knowledge of various support system in mines with its advantages & applicability.

Module 1

Introduction: History of coal mining; coal resource and their geographical distributions; Coalification and factors affecting coalification process, modes of accumulation of coal, evidences in support of in-situ and drift theories; Geological time scale vis -à-vis formation of coal, occurrence and distribution of coal in various stratigraphic horizons; Coal seam structure and abnormalities, geological and other features of Indian coalfields. Bord and Pillar Mining: Choice of methods of mining coal seams; factors affecting choice of mining methods. General principles of Bord and Pillar (B&P) development and associated merits/demerits; Design of B&P workings, statutory provisions related to B&P workings, Semi- mechanized and mechanized schemes of B&P development; Mechanized face loading. Conditions suitable for mechanical loaders and continuous miners.

Module 2

Pillar Extraction: Preparatory arrangement for depillaring operation, statutory provisions on depillaring; principles of designing pillar extraction, factors affecting choice of pillar extraction; partial and full extraction; depillaring with caving and stowing; mechanization in depillaring operation. Local and main fall, indications of roof weighting, measures to bring down roof at regular interval; air blast and measures to minimize its effects; precautions during depillaring operation against fire and inundation; multi-section and contiguous workings. Extraction of pillars in seams prone to bumps.

Module 3

Long wall Mining: Factors affecting long wall mining, long wall face layouts, advancing and retreating faces, single versus double unit long wall faces, orientation of longwall faces; single versus multiple heading gate roads, factors affecting length and width of long wall panel. Extraction of Long wall panel, working with shearer and plough, support system of long wall face and gate roads, case studies of long wall faces in India.

Module 4

Roof Supports: Timber props and cogs; friction/hydraulic props and chocks; other steel supports; types of roof bolts; function, applicability and advantage of roof bolting and cable bolting; powered supports; systematic support rules; supporting scheme of development gallery, B &P and L/W faces, depillaring district; withdrawal of support.

Suggested Reading:

1. *Underground Winning of Coal*, T.N.Singh
2. *Modern Coal Mining Technology*, S.K. Das
3. *Elements of Mining Technology Vol I*, D.J. Deshmukh

Program: B.Tech

Semester: Five

Course: Rock Mechanics

Course Code: 8PCCMiE304

Course Objective:

- ☛ The course provides detailed knowledge on rock properties
- ☛ This will equip the students with the ability to carry out various tests and monitoring the rock behavior.
- ☛ Students will be able in analysis of analysis of data and solving rock mechanics problem in mining and excavation projects.
- ☛ Provides detailed knowledge on rock properties and equips the students with the ability to carry out various tests.
- ☛ Students will be able in analyzing the data and solving rock mechanics problem in mining and excavation projects.
- ☛ Data and solving rock mechanics problem in mining and excavation projects.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the basic knowledge of rock mechanics for understanding, formulating and solving strata control problem in any underground mine.
- ☛ CO2: Explain the various physico mechanical properties of rock in respect to mining.
- ☛ CO3: Acquire knowledge and hands-on competence in applying the concepts of rock failure theories.
- ☛ CO4 :Evaluate the dynamic properties of rock, creep theory and rock mass by various Laboratory test & in situ test

Module 1

Rock mechanics: Definition, history, inherent complexities, source of information and field of application of rock mechanics. Concept of stress and strain in rock: Analysis of stress, strain and constitutive relations in isotropic and anisotropic rocks.

Module 2

Physico-mechanical properties of rock: Determination of physical properties, strengths, strength indices and static elastic constants; Parameters influencing strength; Abrasivity of rock and its determination.

Module 3

Failure criteria for rock and rock mass: Theories of rock failure; Coulomb, Mohr and Griffith criteria; Empirical criteria. Pre-mining state of stress: Sources, methods of determination including over coring, hydro-fracturing methods and other methods.

Module 4

Dynamic properties of rock and rock mass: Propagation of elastic wave in rock media; Determination of dynamic strength and elastic constants of rock. Time dependent properties of rock: Creep deformation and strength behavior; Creep test and rheological models. Strength and Deformability of Rock Mass: In situ shear tests; Evaluation of shear strength; In situ bearing



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strength test; In situ deformability tests-Plate Loading Test, Plate Jacking Test and Bore hole Jack Tests.

Suggested Reading:

1. *Elements of Mining Technology Vol I, D.J. Deshmukh*
2. *The elements of mechanics of mining ground, B.S. Verma*
3. *Rock Mechanics for Engineers, Dr. B.P. Verma*



Program: B.Tech

Semester: Five

Course: Rock Mechanics Lab

Course Code: 8PCCMiE304P

List of Experiments:

1. Preparation of rock sample for testing in laboratory.
2. Methods for determination of compressive strength, tensile strength, shear strength and triaxial strength of rock.
3. Porosity of rock.
4. Abrasivity of rock.
5. Strength indices of rock.
6. Modulus of elasticity and Poisson's ratio.
7. Slake durability of rock.
8. Determination of in situ stresses in rock.

Program: B.Tech

Semester: Five

Course: Numerical and Statistical Method

Course Code: 8PCCMiE305

Course Objective:

- ☛ The students will get the concept about finite element models, methods and boundary elements method and its practical applications in mining and rock mechanics.
- ☛ To understand the practical applications of numerical methods in mining field.
- ☛ The understanding of the mathematical principles on numerical differentiation and integration and numerical solutions to ODE would provide them the ability to formulate and solve some of the physical problems of engineering.
- ☛ To acquaint the student with analysis of correlation and Eigen value problems used in wide variety of situations.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: - Apply numerical methods to find solutions of algebraic and transcendental equations using different methods under different conditions
- ☛ CO 2: - Apply various interpolation methods and finite difference concepts to solve practical problems of Numerical Integration.
- ☛ CO 3: - To give a brief idea about advance probability concepts for solving practical problems of real world.
- ☛ CO 4: - Apply statistical tools like correlation, regression to solve various engineering problems.

Module 1

Numerical Methods: Solution of algebraic and transcendental equations by bisection, iteration, false position and Newton Raphson methods. Solution of a system of linear simultaneous equations by Gauss elimination, Gauss-Jordan, Jacobi and Gauss Seidel methods.

Module 2

Finite difference, Symbolic relations, Interpolation and extra polation, Newton-Gregory forward and backward, Gauss forward and backward, Numerical differentiation and integration, Trapezoidal, Simpson's $1/3^{\text{rd}}$, Simpson's $3/8^{\text{th}}$, Euler's, Modified Euler's, Runge-Kutta and Milne's methods.

Module 3

Probability: Various approaches of probability, two theorems (without proof), conditional probability, Bayes Theorem. Random variable: Definition, probability mass & density functions, distribution function, mathematical expectation and moment generating function.

Module 4

Measure of central tendency, Measure of Dispersion. Statistical Methods: Moments, skewness and kurtosis; curve fitting. Correlation and Regression: correlation coefficients, regression lines, regression coefficients and their properties.

Suggested Reading:

1. Numerical & Statistical method, D.R. Vittal
2. Numerical Methods in Engineering and Science" by Grewal B.S. and Grewal J.S

Program: B.Tech

Semester: Five

Course: Remote Sensing & GIS

Course Code: 8PCCMiE306

Course Objective:

- ☛ This course enables students to create a map that accurately shows distances, areas, or directions.
- ☛ Applications of remote sensing and GIS in geological mapping and mineral exploration.
- ☛ They will have deep knowledge about geophysics, remote sensing and GIS.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the basic knowledge of remote sensing & concept of EMR.
- ☛ CO2: Apply the knowledge of Aerial photography and photogrammetry.-electromagnetic Radiation, Remote Sensing Data Product, Spatial Filtering, Band Ratios, Image Classification GIS-Project Planning, Management and Implementation.
- ☛ CO3: Learners know about Basics of GIS -components of GIS, History, Geospatial data architecture, Operations, Geographic coordinate systems, Map projections etc.
- ☛ CO4: Analyze the vector data & processing and Applying the use of GIS.

Module I – REMOTE SENSING -INTRODUCTION AND CONCEPTS (9 hours)

Introduction of Remote Sensing – Energy sources and Radiation principles, Energy equation, EMR and Spectrum, EMR interaction with Atmosphere- scattering, Absorption, EMR interaction with earth surface features- reflection, absorption, emission and transmission, Spectral response pattern, vegetation, soil, water bodies- Spectral reflectance.

Module II - AERIAL PHOTOGRAPHY AND PHOTOGRAMMETRY (9 hours)

Introduction-, Terrestrial and Aerial photographs - vertical and oblique photographs - height determination, contouring - photographic interpretations - stereoscopy – parallax bar- Flight Planning- Photo Interpretation, Applications of aerial Photos-Photo theodolite.

Module III - BASICS OF GIS (9 hours)

Introduction, concepts, Information system, components of GIS, History, Geospatial data architecture, Operations, Geographic coordinate systems, Map projections, concepts, Input data for GIS, display, types of output products. GIS categories, Level and scale of Measurement, importance of data quality.

Module IV - VECTOR DATA & PROCESSING (9 hours)

GIS data types, data Representation, Data sources, typical GIS data sets, Data Acquisition, vector data model, relationship between classes, data structure, Database Management, Data Management and Analysis; Advantages and disadvantages of Raster or Vector Data; Uses of GIS.

Suggested Reading:

1. *Surveying and Levelling – R. Agor*
2. *Higher Surveying – B C Punamia*
3. *Surveying and Levelling – Basak*

Program: B.Tech

Semester: Five

Course: Vocational Trainee Report II

Course Code: 8PROJMiE301

Course Objective:

- ☛ To provide training in mines for gaining thorough understanding of all the theoretical knowledge.
- ☛ Gaining practical experience is an important aspect of the mining engineering program having many characteristic features of its own.
- ☛ The students will have insight about mining methods and techniques.
- ☛ The outcome at the place of work is always much more than what can be learned in the class room.
- ☛ To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.
- ☛ To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.

Course Outcome:

After the completion of these topics the student should be able to

- ☛ Mining graduates would be equipped with managerial skill which would be useful to them for achieving their program educational outputs.

1. Study of History of Mine – Note name of the Owner, Agent, Manager, SafetyOffice
2. Study of Mine geological information
3. Study of Mine Plans and Sections
4. Study of Surface features related to Mine
5. Study of method of working
6. Study of method of blasting
7. Study of Transportation system and layouts
8. Study of Ventilation systems and layouts
9. Study of Drainage system
10. Study of Pit top and Pit bottom layouts.
11. Study of man Power plan
12. Develop the Lamp room layout and Magazine Layout
13. Draw the charts depicting instructional items related to Mining subjects

Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of Semesters.

Program: B.Tech

Semester: Five

Course: Constitution of India

Course Code: MC301

Course Objective:

The basic object of the course is to provide the acquaintance with the basic features of Indian Constitution e.g. Fundamental Rights, Fundamental Duties and Directive Principles of State policy. Further it is aimed to impart the knowledge about Judicial system in India.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the basic features of our constitution.
- ☛ CO2: Learn the concept of Right to Equality & Prohibition on grounds of Religion, Race, Caste, sex & Place of Birth.
- ☛ CO3: Understand the concept of Basic freedoms, Protection in respect of conviction for offences & Right to Life and Personal Liberty.
- ☛ CO4: Explain the Right against exploitation & Freedom of Religion.
- ☛ CO5: Understand the concept of Directive Principles of State Policy and their relation with Fundamental Rights, Fundamental Duties & Right to Property.

Module-I (4 Hours)

1. Nature of Indian Constitution
2. Preamble
3. Union and its Territory (Arts 1-4)
4. Citizenship (Arts 5-11)
5. Definition of State (Art 12)

Module-II (6 Hours)

1. Judicial Review (Article 13)
2. Right to Equality (Article 14)
3. Prohibition on grounds of Religion, Race, Caste, Sex, Place of Birth (Article 15)
4. Equality of Opportunity in Public Employment (Article 16)
5. Abolition of Untouchability and Titles (Articles 17-18)

Module-III (6 Hours)

1. Basic freedoms (Article 19)
2. Protection in respect of conviction for offences (Article 20)
3. Right to Life and Personal Liberty (Article 21)
4. Safeguards against arbitrary arrest and detention (Article 22)

Module-IV (6 Hours)

1. Right against exploitation (Articles 23-24)
2. Freedom of Religion (Articles 25-28)
3. Cultural and Educational Rights of Minorities (Articles 29-30)
4. Constitutional Remedies (Articles 32-35)

Module-V (8 Hours)

1. Directive Principles of State Policy and their relation with Fundamental Rights (Articles 36-51)
2. Fundamental Duties (Article 51-A)
3. Right to Property (Article 300-A)
4. Facts and Law laid down in Maneka Gandhi V Union of India AIR 1978 SC597
5. Facts and Law laid down in Indira Sawhney V Union of India AIR 1993SC 477

Suggested Reading:

1. *Austin Granville- The Indian constitution: Cornerstone of a Nation.*
2. *Seervai H.M. - Constitution of India*
3. *Jain M.P. – Indian Constitutional Law*
4. *Shukla V N- Constitution of India (ed. By M.P. Singh)*
5. *Basu D.D. – Shorter Constitution of India*

Program: B.Tech

Semester: Five

Course: Professional Skills

Course Code: HSMC301

Course Objective:

The objectives of the course are:

- ☛ To make the student understand the role of communication of personal & professional success.
- ☛ To develop awareness of appropriate communication strategies.
- ☛ To prepare and present messages with a specific intent.
- ☛ To analyze a variety of communication acts.
- ☛ To ethically use, document and integrate sources.
- ☛ To create a basic awareness about the significance of soft skills in professional and inter-personal communication.
- ☛ To facilitate an overall development of the personality.

Course Outcome:

At the end of the course learners will be able to:

- ☛ CO 1: Identify and coordinate life and professional goals.
- ☛ CO 2: Learn self-improvement through interpersonal skills.
- ☛ CO 3: Apply solutions to challenges.
- ☛ CO 4: Adapt to changes.
- ☛ CO 5: Manage life and work balance.

Unit I: Personal Development

Managing Self - Self Discovery, Self Awareness, Self Esteem, Self Responsibility, Self Management

Personal Development Skills, Categories of Personal Development, Personal Development Process

Relationship Management - Managing Others, Interpersonal Skills, Improving Relationship, Transactional Analysis, JOHARI Window, four Life Positions

Unit II: Thinking Process

Strategic Thinking – Introduction, Concept, Stages in Strategic Thinking, Process of Strategic Thinking, Importance of Strategic Thinking, Characteristics of Strategic Thinkers, Developing Strategic Thinking

Lateral Thinking – Introduction, Meaning, Need for Lateral Thinking, Techniques of Lateral Thinking, Benefits of Lateral Thinking

Creative Thinking – Out of Box Thinking, Application of Thinking

Facing Changes – Adapting Change, Understanding Change- Examples of Organizational Change

Facing Challenges – Introduction, Taking Initiative, Benefits of facing challenges, Facing challenges in life

Balancing Work and Life – Importance, Gender differences regarding work life balance, Tips for balancing work and life



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Unit III: Individual Behavior

Attitude – Components of Attitude, Factors influencing Attitude, Types of Attitude, Challenges and lessons from Attitude, Impact of Attitude on Behaviour

Motivation – Concept, Objective, Factors of Motivation, Self Steem, Intrinsic & Extrinsic Motivation

Time Management – Value of Time, Diagnosing Time management, Weekly Planner, To Do List, Prioritizing Work

Stress Management – Introduction, Difference between Stress, Anxiety and Tension, Managing Stress

Applied Ethics – Introducing Professional Ethics, Ethical Dilemma

Unit IV: Employment Communication

Job Communication – Developing Job Communication Skills, Job Communication Process, Developing Confidence; Job Search Strategy – Understanding the Job Market, The Job Search Process, Job Search Techniques; Job Application, Employment Letters

Resume Building – Difference between Bio data, Curriculum Vitae and Resume

The Job Interview - Types of Job Interview, Preparing for a Job Interview, Understanding Interview Questions, Handling Interview Questions, Interview Strategies

Psychometric Test

Suggested Readings:

1. Covey S (2004) *The 7 Habits of Highly Effective People*.
2. Goud, N. & Arkoff, A. (2003) *Psychology and Personal Growth*, Allyn & Bacon.
3. Sen, Leena, *Communication Skills*, Eastern Economy Edition
4. Dr. K.Alex *Managerial Skills*, S.Chand

MINING ENGINEERING

B.Tech



SYLLABUS



RKDF UNIVERSITY RANCHI

(2022-2026)

SEMESTER VI



RKDF UNIVERSITY RANCHI

B.TECH SEMESTER VI

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER VI					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Professional Core Courses	8PCCMiE308	Underground Metal Mining	100	3
2	Professional Core Courses	8PCCMiE309	Advanced Underground Mining	100	3
3	Professional Core Courses	8PCCMiE310	Mine Ventilation	100	3
4	Professional Core Courses	8PCCMiE311	Mine Planning	100	3
5	Professional Core Courses	8PCCMiE312	Ground Control	100	3
6	Humanities and Social Sciences	HSMC302	**Soft Skills & Interpersonal Communication	100	0
PRACTICAL / SESSIONAL					
1	Professional Core Courses	8PCCMiE310P	Mine Ventilation Lab	50	1
2	Professional Core Courses	8PCCMiE312P	Ground Control Lab	50	1
3	Professional Core Courses	8PCCMiE313P	Mine Planning & Design Exercise	50	1
			TOTAL	750	18
**NOTE: Qualifying Non Credit Course.					

Program: B.Tech

Semester: Six

Course: Underground Metal Mining

Course Code: 8PCCMiE308

Course Objective:

- ☛ The students will have good knowledge about the various advanced methods of metal mining and special mining techniques to overcome the field issues.
- ☛ The students will have basic concept on metal mining methods, mine design, development and operations of metal mines. They will also know about novel methods of metal mining and its applications.
- ☛ To understand the various advanced and recent methods of metal mining.
- ☛ Enables the students to select suitable methods of working underground metal mines and decide the necessary parameters of mine construction.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Develop the understanding of metal mining industry.
- ☛ CO 2: Sketch the development of a working underground metal mines.
- ☛ CO 3: Acquire knowledge and hands-on competence in applying the concepts in the design of metal mine.
- ☛ CO 4: Recommend the various support system for underground metal mine working.

Module 1

Introduction: Present status of Indian metal mining industry; Scope and limitations of underground mining.

Module 2

Development: Choice of level interval and back/block length; Shape, size, position, excavation and equipping of shaft station/plat, grizzly, ore/waste bin, main ore pass system, underground crushing and loading stations, underground chambers, sump and other subsidiary excavations; Arrangements for dumping into main ore pass; Cross-cuts and drifts their shape, size and position; Review of excavation process groundbreaking, mucking, ventilation and support; Track extension and car switching; Use of modern drilling and loading equipment in drifting; Raises and winzes their shape, size and position; Excavation process ground breaking, mucking, ventilation and support ; Modern methods of raising-Alimakand Jora- lift raising, long hole method including vertical crater retreat method of raising; Raise boring systems and their details; Modern methods of winzing; Secondary breaking at grizzlyconventional and mechanized methods.

Module 3

Stoping : Selection of stopingmethods; Classification of stoping methods; Stoppingof narrow ore bodies by underhand, overhand, breast, long hole and raise mining methods; Resuing; Mining of parallel veins; Room & pillar, sublevel, large diameter blast hole/DTH, cascade, shrinkage and vertical crater retreat methods, their applicability, stope layouts ,stope preparation, ground breaking, mucking, ventilation and supporting; Haulage and dumping; Supported methods– horizontal over hand and under hand cut-and-fill methods, square-set method and its variations, details of, ground breaking, supporting, mucking, ventilation, haulage and dumping.

Module 4

Mine supports: Timber support: Post, drift-set of various types, square-set, crib-set, cog, stull and chock/chock mat supports; fore poling/piling; load bearing capacity of timber supports; bulkheads. Steel support: Steel set-rigid and yielding types; tubbing, wire mesh, steel lining,; improvised steel props, friction props, hydraulic props; link bars and chocks, powered supports; Cement support: Poured monolithic and reinforced concrete lining; monolithic pump packing, concrete blocks, concrete slabs, gunitingand shotcreting. Rock support: Pillars of ore and waste, packwalls, masonry walls and arches building materials and construction.

Suggested Reading:

1. *Elements of Mining Technology Vol I, II, D.J. Deshmukh, Central techno publication, Nagpur*
2. *Surface Mining, G.B. Misra, Oxford University Press, Calcutta*
3. *Introductory Mining Engineering, H.L. Hartman, A Wiley Interscience Publication, Singapore*

Program: B.Tech

Semester: Six

Course: Advanced Underground Mining

Course Code: 8PCCMiE309

Course Objective:

- ☛ To introduce the recent trends of level of mechanization for coal face.
- ☛ To understand the various advanced methods of coal mining.
- ☛ To study the various special methods of winning coal.
- ☛ To study the development of panels and extraction of coal in Bord and Pillar method.
- ☛ The students will have knowledge on development and depillaring of coal in Bord and Pillar and advancing and retreating in Longwall methods. They will also know about novel methods of winning of coal.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Define & explain the underground mining of thick seam & hydraulic mining.
- ☛ CO 2: Analyze the design of thin seam mining method in underground & coal gasification techniques.
- ☛ CO 3: Describe the working methods & present status of underground metal mining.
- ☛ CO 4: Design & solve the deep mining method & special mining methods.

Module 1

Underground Coal Mining of thick seams: Concept of a thick seam; Problems of mining thick seams; Past experience of working thick seams by Bord & Pillar method in multi-sections; Longwall based multi-slice methods:- Inclined slicing, horizontal slicing and cross-slicing in ascending and descending sequence; Under winning methods:-sub-level caving, integral caving, blasting gallery method, descending shield method.

Hydraulic Mining of Coal: Conditions suitable for hydraulic Mining of Coal, Hydraulic Mining Operation, Hydraulic breaking of coal, hydraulic transport and hydro hoisting; Layout of workings for hydraulic mining of moderately thick seams.

Module 2

Mining of thin seams: Problems in Mining thin seams; Equipment and methods for thin seam extraction. Underground Coal Gasification: Conditions suitable for Underground Coal Gasification; Basic principle and technology of underground coal gasification, Advantage and disadvantage of UCG: Scope of application of UCG in Indian conditions, Highwall Mining: Introduction, Applicability and Method

Module 3

Underground Metal Mining Introduction: Current status of recent developments in underground metal mining methods in India and abroad. Caving methods: Top slicing, sub-level caving and block caving methods; Stope layouts, stope preparation and production operations; Design and construction of draw points; Mechanics of draw and draw control procedure; Recovery and dilution. Combined systems: Combined open-room, shrinkage, and cut-and-fill systems; combined systems with subsequent filling of rooms.

Module 4

Deep mining: Problems of deep mining and the remedial measures; Design and layout of stopes in rock burst prone mines. Special methods: Hydraulic, thermal, hydro chemical and biochemical

methods; Underwater/sea-bed mining- current status; recent developments in underground metal mining.

Suggested Reading:

1. *Elements of Mining Technology Vol I, II, D.J. Deshmukh, Central Techno Publication, Nagpur*
2. *Surface Mining, G.B. Misra, Oxford University Press, Calcutta*
3. *Introductory Mining Engineering, H.L. Hartman, A Wiley – Interscience Publication, Singapore.*

Program: B.Tech
Semester: Six
Course: Mine Ventilation
Course Code: 8PCCMiE310

Course Objective:

- ☛ Enables the students to understand the airflow in sufficient quantity and quality to dilute contaminants to safe concentrations in all parts of the facility where personnel are required to work or travel.
- ☛ Enables the students to understand the ventilation requirements.
- ☛ Students will be able in selecting appropriate mine fans and design suitable ventilation structures.
- ☛ The students will have deep knowledge about the mine accidents, disaster, disease and mine safety with risk assessment, mitigation and management.
- ☛ To learn ventilation methods and strata monitoring instruments.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Classify mine gases & summaries various ventilation instruments.
- ☛ CO 2: Illustrate the technical knowledge on origin, occurrence, effects, and detection of various mine gases, air conditioning of surface and underground mining.
- ☛ CO 3: Explain & differentiate between natural ventilation & mechanical ventilation.
- ☛ CO 4: Design ventilation plan & layout considering various parameters.

Module 1

Composition of mine atmosphere: Mine gases □ production, properties and effects; Sampling and analysis of mine air; Methane content; Methane drainage; Flame safety lamp and its uses; Methanometer; Methane layering; Radon gas and its daughter products; Monitoring of gases.

Module 2

Heat and humidity: Sources of heat in mines; Effects of heat and humidity; Psychrometry, Kata thermometer; Air-conditioning. Air flow through mine openings: Laws of flow, resistance of airways, equivalent orifice, losses in airways, distribution of air, economic design of airways; Flow control devices; Permissible air velocities in different types of workings/openings; Standards of ventilation.

Module 3

Natural ventilation: Causes, effect of seasonal variations, calculation of NVP from air densities, thermodynamic principles and other methods. Mechanical ventilation: Types of mine fans; Theory, characteristics and suitability of fans; Selection, testing and output control; Fans in series and parallel; Forcing and exhaust configurations; Reversal of flow; Fan drifts, diffusers, evasee; Booster and auxiliary ventilation; Venturi blowers; Ventilation of deep mines □ underground and open pit.

Module 4

Ventilation planning: Planning of ventilation systems and economic considerations; Ventilation layouts for mining of coal and ore deposits; Ventilation of workings/stopes using heavy blasting; Calculation of air quantity required for ventilating a mine; Calculation of total mine head; Network analysis principles and computer applications; Automation and remote control of ventilation installations; Ventilation surveys.

Suggested Reading:

1. *Elements of Mining Technology Vol, II, D.J. Deshmukh, Central techno publication, Nagpur*
2. *Mine Environment Ventilation, G.B. Misra, Oxford University Press, Calcutta*
3. *Mine Disaster & Mine Rescue, M.A. Ramlu, Oxford University Press, Calcutta*

Program: B.Tech
Semester: Six
Course: Mine Ventilation Lab
Course Code: 8PCCMiE310P

List of Experiments:

1. Study and use of MSA-D6 Methanometer.
2. Flame safety lamp—construction and gas testing.
3. Detection of carbon monoxide using detector tube.
4. Analysis of mine gases using Gas Chromatograph.
5. Measurement of air flow using velometer, vane anemometer and Pitot tube.
6. Plotting off a characteristic curves.
7. Determination of psychrometric properties of air.
8. Estimation of air cooling power using Kata thermometer.

Program: B.Tech
Semester: Six
Course: Mine Planning
Course Code: 8PCCMiE311

Course Objective:

- ☛ To introduce the various techniques for mine planning, geotechnical investigation and equipment management.
- ☛ To appreciate the modern trends in opencast mines, safety and environment.
- ☛ The students will have knowledge about the available mineral resources, exploration techniques and its strategy. They will know about the methods of preparation of feasibility reports and its evaluation techniques.

Course Objective:

After the completion of the course, student will be able to:

- ☛ CO 1: Apply knowledge of mine planning for understanding, formulating and solving mine planning & scheduling problems.
- ☛ CO 2: Describe stripping ratio and opencast working & geometry.
- ☛ CO 3: Acquire knowledge and hands-on competence in applying the concepts in the development of mine planning.
- ☛ CO4: Analyze feasibility & detailed report of opencast mining.

Module 1

Basic Concepts: Objectives of mine planning. Characteristics of planning process. Planning stages – long, intermediate and short range planning. Technical, economical and environmental information required for mine planning. Preparation of Feasibility and Project Report

Module 2

Stripping ratio Concept of stripping ratio; Types of stripping ratios and their significance; Choice between surface and underground mining. Geometrical considerations Basic bench geometry; Ore access; Pit slope geometry; Addition of haul road on pit plan; Pit layouts.

Module 3

Underground Coal Mine Planning: Mine planning components and planning steps; Optimization of mine size (mine production capacity) based on techno-economic considerations; Mine system and sub-systems; Equipment and face scheduling against targeted production: Closure of underground mines.

Module 4

Feasibility Report and Detailed Project Report: Contents, preparation and economic evaluation.

Suggested Reading:

1. *Elements of Mining Technology Vol I, II, D.J. Deshmukh, Central techno publication, Nagpur*
2. *Surface Mining, G.B. Misra, Oxford University Press, Calcutta*
3. *Introductory Mining Engineering, H.L. Hartman, A Wiley- Interscience Publication, Singapore.*
4. *Principles of Mine Planning, Jay ant Bhattacharjee*

Program: B.Tech

Semester: Six

Course: Ground Control

Course Code: 8PCCMiE312

Course Objective:

- ☛ The students will have knowledge on rock mechanics instrumentation, approach to pit slope stability, theories of subsidence and failure of rocks.
- ☛ The students will have knowledge on time dependent properties of rock, subsidence, ground vibration monitoring, stowing characteristics and permeability.
- ☛ They will also know about design of underground opening and methods of stowing.
- ☛ Student will know about subsidence control, governing norms and regulations.
- ☛ The students will know the fundamental mechanics of rock slope failure, types of failure and its influencing parameters.
- ☛ The students will have the concept about the rock mass classification, mechanism of rock reinforcement, existing and special methods of rock reinforcement.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Sketch of strata control for understanding, formulating and solving strata control problem in any underground mine.
- ☛ CO 2: Analyze and solve strata movement problems.
- ☛ CO 3: Acquire knowledge and hands-on competence in applying the concepts in the development of strata control.
- ☛ CO 4: Describe subsidence & develop in depth knowledge of monitoring prediction & control.

Module 1

Engineering classification of rocks and rock masses: Classification systems in rock engineering; Classification of intact rocks; Classification of rockmasses- Terzaghi's rock load, RQD, Rock Structure Rating, Bieniawski's RMR, Barton's Q-System Rock mass classification and Recent developments; correlations between different classification systems; Applications of Rock mass Classification in rock engineering.

Module 2

Design of mine pillars: Mine pillars and their classification; pillar mechanics; Design of mine pillars and shaft pillar: stresses acting on pillars; stress distribution in pillars; mechanics of pillar failure; interaction of pillar, floor and roof; design of rooms and pillars; design of barrier and yield pillars. Design of support for underground excavation:

Module 3

Rock support: principles of supports; Types and classification of support; support design methodology; support design criteria; rock- support interaction analysis; support requirement under different conditions – influencing parameters, selection and estimation. Design of rock reinforcement for underground excavation: Development of reinforcement by rock bolting; rock bolt types; theories of rock bolting; design of rock reinforcement system.

Module 4

Subsidence: Causes and impacts of subsidence; Mechanics of surface subsidence, discontinuous and continuous subsidence; Monitoring, prediction, control and management of subsidence. Rock burst and coal bump: Phenomenology of rock bursts and coal bump; causes prediction, monitoring.

Suggested Reading:

1. *Elements of Mining Technology Vol, I, D.J. Deshmukh, Central techno publication, Nagpur*
2. *Modern Coal Mining Technology, S.K. Das, Mintech publication Bhubaneshwar*
3. *The Elements of Mechanics of Mining Ground, B.S. Verma, Tuhin& Co. Lucknow*

Program: B.Tech

Semester: Six

Course: Ground Control Lab

Course Code: 8PCCMiE312P

List of Experiments:

1. Load-yield characteristics of friction prop and hydraulic prop/leg.
2. Drillability of rock.
3. Anchor age strength of rock bolt.
4. Rock fracture toughness.
5. Dynamic modulus of elasticity.
6. Shear properties of discontinuity.
7. Blast induced ground vibration.
8. Use of high speed video camera.
9. Use of rock mass monitoring system in assessing blasting performance.
10. Load cell, extensometer, vibrating wire stress meter and convergence meter.

Program: B.Tech
Semester: Six
Course: Mine Planning & Design Exercise
Course Code: 8PCCMiE313P

Course Objective:

- ☛ The students will have knowledge on planning of opencast mining, underground mining and equipment utilization.
- ☛ To understand the planning of opencast mining, underground mining and equipment Utilization.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Design layout of opencast mine to know the calculation of output.
- ☛ CO 2: Plan layout of opencast mine to know organizations charts for mine management.
- ☛ CO 3: Analyze & draw layout of with a target output.
- ☛ CO 4: Design layout of opencast mining with equipment scheduling.

List of Experiments:

1. Know the calculation of output for open cast mine

Calculation of No. of shovels, dumpers, drills require for given output. Calculation of bucket capacity of the shovel, drill capacity, capacity of the dumper for the above problem.

2. Know the organizations charts for mine management.

Draw the Organizations charts for one Mechanized underground Coal Mine.

3. Draw a layout for a surface mine with an output 10,000 tonne/day in a flat deposit.

Draw a Layout for Mechanized opencast mine assume require condition data-Design of blasting sequence in surface mine.

4. Design of Open cast Coal Mines.

5. Design of underground Coal & Metalliferous mines including entries, ventilation, support, transport system etc.

Program: B.Tech
Semester: Six
Course: Soft Skills & Interpersonal Communication
Course Code: HSMC302

Course Objective:

The objectives of the course are:

- ☛ To develop inter personal skills and be an effective goal oriented team player.
- ☛ To develop professionals with idealistic, practical and moral values.
- ☛ To develop communication and problem solving skills.
- ☛ To re-engineer attitude and understand its influence on behavior.
- ☛ To encourage the all round development of students by focusing on soft skills.
- ☛ To make the engineering students aware of the importance, the role and content of soft skills through instruction, knowledge acquisition, demonstration.

Course Outcome:

At the end of the course learners will be able to:

- ☛ CO 1: Understand the importance of Soft Skills and Interpersonal Communication in Professional world.
- ☛ CO 2: Effectively communicate through verbal /oral communication and improve the listening skills.
- ☛ CO 3: Become more effective individual through goal/target setting, self motivation and practicing creative thinking.
- ☛ C O 4: Function effectively in multi-disciplinary and heterogeneous teams through the knowledge of team work, inter –personal relationships, conflict management and leadership quality.

Unit I: Self Analysis

- Introduction to Soft Skills and Hard Skills, Importance of Soft Skills, Attributes regarded as Soft Skills, Identifying and improving your Soft Skills, Art of Negotiation
- Stage Fright
- Self Discovery, Importance of knowing oneself, Process of knowing oneself, SWOT Analysis, Benefits of SWOT analysis, SWOT Analysis , Self Esteem, Ways to improve Self Esteem, Aristotle on Self-Gender and Self, Feminist Self, Escaping the Self
- Self Development
- Developing Positive Attitude and Self Confidence, Forming Values

Unit II: Goal Setting and Career Planning

- Wish List, SMART Goals, Blue print for Success, Short term, Long Term, Life Time Goals,
- Art of Listening, Art of Reading, Art of Speaking, Art of Writing, writing E-mail
- Motivation Skills, Personality Development, Improving Perception
- Time Management, Stress Management, Conflict Handling
- Problem Solving and Decision Making, adaptability.

Unit III: Effective Communication

- Communication Skills, Concept/Meaning, Definition
- Types of Communication, Process of Communication, stages of Communication



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- Barriers to Communication
 - Communication Network
 - 7 C's of Communication
- Verbal & non verbal Communication

Unit IV: Interpersonal Skills

- Interpersonal Communication, Basic Skill set
- Effective Interpersonal Communication in Organization
- Team Building, Communicating in a Team
- Intercultural Communication
- Leadership traits through Communication
- Communicating assertively
- Presentation Skills

Suggested Readings:

1. Covey Sean *Seven Habits of Highly Effective Teens*, New York, Fireside Publishers, 1998.
2. Carnegie Dale, *How to win Friends and Influence People*, New York: Simon & Schuster, 1998.
3. Thomas A Harris, *I am ok, You are ok*, New York-Harper and Row, 1972
4. Dr. K. Alex *Soft Skills*, S. Chand

MINING ENGINEERING

B.Tech



SYLLABUS (2022-2026) SEMESTER VII



B.TECH SEMESTER VII

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER VII					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Professional Core Courses	8PCCMiE401	Mine Environmental Engineering	100	3
2	Professional Core Courses	8PCCMiE402	Environmental Aspect of Mining	100	3
3	Professional Core Courses	8PCCMiE403	Mining Machinery II	100	3
4	Professional Core Courses	8PCCMiE404	Mine Legislation & Safety I	100	3
5	PROFESSIONAL ELECTIVE (Any one of the following subjects)				
	Professional Elective Courses	8PECMiEEL401	Dimensional Stone Mining	100	3
	Professional Elective Courses	8PECMiEEL402	Rock Excavation Engineering		
	Professional Elective Courses	8PECMiEEL403	Design of Open Pit Mines		
6	Humanities and Social Sciences	HSMC401	**Seminar in Executive Communication	50	0
PRATICAL / SESSIONAL					
1	Professional Core Courses	PCCMiE401P	Mine Environmental Engineering Lab	50	1
2	Project	PROJMiE401	Vocational Trainee Report II	100	2
			TOTAL	700	18
**NOTE: Qualifying Non Credit Course.					

Program: B.Tech

Semester: Seven

Course: Mine Environmental Engineering

Course Code: 8PCCMiE401

Course Objective:

- ☛ The students will have knowledge on spontaneous heating, mine fires, inundation and explosions.
- ☛ To study about mine rescue and first aid.
- ☛ The students will get practical knowledge about underground mine ventilation methods and planning.
- ☛ The students will have deep knowledge about the mine accidents, disaster, disease and mine Safety with risk assessment, mitigation and management.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the basic concept of mine fire & spontaneous combustion with detection, control, dealing and prevention in underground mine.
- ☛ CO2: Explain the causes & prevention of mine explosions like firedamp & coal dust explosion.
- ☛ CO3: Gain knowledge of causes & prevention of inundation and uses of the rescue apparatus properly.
- ☛ CO4: Analyze the causes of production of air born dust and its control as well as prevention and the standard of illumination at working place and other parts of the mine.

Module 1

Mine fires: Causes and classification of mine fires; Spontaneous combustion mechanism, stages of spontaneous combustion, susceptibility indices, factors affecting spontaneous combustion; Detection and prevention of spontaneous heating and accidental fires; Dealing with mine fires direct and indirect methods, fire stopping; Re-opening of sealed-off areas; Fires in quarries, Coal stacks and waste dumps.

Module 2

Mine explosions: Firedamp and coal dust explosions causes and prevention, explosive limits; Stone- dust and water barriers; Explosion in quarries over developed pillars; Investigation after an explosion.

Module 3

Inundation: Causes and prevention; Precautions and techniques of approaching old workings; Dewatering of waterlogged working, safety boring apparatus, pattern of holes; Design and construction of water dams. Rescue and recovery: Rescue equipment and their uses, classification of rescue apparatus; Resuscitation; Rescue stations and rescue rooms; Organization of rescue work; Emergency preparedness and response system.

Module 4

Air borne respirable dust: Generation, dispersion, measurement and control; Physiological effects of dust, dust-related diseases. Illumination: Cap lamps; Layout and organization of lamp rooms; Standards of illumination; Photometry and illumination survey; Lighting from main and other sources.

Suggested Reading:

1. *Elements of Mining Technology Vol II*, D.J.Deshmukh, Central Techno Publication, Nagpur
2. *Mine Environment & Ventilation*, G.B.Misra, Oxford University Press, Calcutta
3. *Mine Disaster & Mine Rescue*, M.A. Ramlu, Oxford University Press, Calcutta



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Program: B.Tech

Semester: Seven

Course: Mine Environmental Engineering Lab

Course Code: 8PCCMiE401P

List of Experiment:

1. Detection of Methane using flame safety lamp.
2. Demonstration of whirling hygrometer and determination of relative humidity using whirling hygrometer.
3. Demonstration of Kata thermometer and determination of cooling power by Kata thermometer.
4. Demonstration of various ventilation devices.
5. Demonstration of vane Anemometer and determination of quantity by Anemometer.
6. Demonstration of velometer and measurement of air velocity by velometer.

Program: B.Tech

Semester: Seven

Course: Environmental Aspects of Mining

Course Code: 8PCCMiE402

Course Objective:

- ☛ To understand the sustainable development & environmental laws.
- ☛ To develop knowledge of water pollution due to mining activities and its effect & treatment.
- ☛ Student will be able to understand the air pollutions due to mining activities and its effect & treatment.
- ☛ Students will be able to concept of Corporate Social Responsibility & Mine closure.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the sustainable development and environmental as well as ecological conservation along with mining activities.
- ☛ CO2: Learn the composition of mine and atmospheric air and the source of pollutants of air, water and sound.
- ☛ CO3: Apply the knowledge on control of pollution of water and its treatment for purification using important parameters such as-BOD, COD and DO.
- ☛ CO4: Apply the knowledge on environmental impact assessment of the project and laws related to land acquisition and rehabilitation as well as mine closure and corporate social responsibility.

Module 1

Introduction: Sustainable development, environmental carrying capacity - concepts & principles; Environmental impacts of mining and associated activities. Ecology: Introduction to ecology, ecosystem structures and functions.

Module 2

Air pollution: Atmospheric composition and meteorology; Sources of air pollution—point and non-point; Emission factors; Control measures— extraction, suppression and consolidation of dust. Noise and vibration: Basic concepts, sources, monitoring and control measures.

Module 3

Water pollution: Global hydrological cycle; Self-purification mechanism, sources of water pollution, important parameters—pH, turbidity, oil & grease, nitrates, DO, BOD, COD; Eutrophication, de oxygenation, acid mine drainage and heavy metal pollution— preventive and control measures. Land environment: Land degradation due to mining; Physical and biological reclamation.

Module 4

Environmental administration: Laws related to mining environment; EIA of mining projects. Land Acquisition & Revenue: Concepts: Related laws and regulations. Corporate Social Responsibility: Concepts and principles. Mine closure: Concepts and principles.

Suggested Reading:

1. *Mine Environment & Ventilation*, G.B.Misra, Oxford University Press, Calcutta
2. *Environmental Impact of Mining and Mineral Processing*, Ravi Jain
3. *Hand Book of Methods in Environment Studies*, S.K.Maiti, ISM
4. *Environment Management In Mining Area*, Dr.N.C.Saxena

Program: B.Tech

Semester: Seven

Course: Mining Machinery II

Course Code: 8PCCMiE403

Course Objective:

- ☛ The students will get exposure towards the material handling methods and systems and its principle to convey the minerals or materials from mines, plants and workshops.
- ☛ The students will have practical knowledge about underground machineries, ropes, pit-top and pit-bottom layouts.
- ☛ Enables students about the conveyor system and its advancement.
- ☛ To understand riding system and aerial system

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Learn the basic concept of mine machinery for selecting the suitable machines and solving the excavating, loading & transportation problems in mine.
- ☛ CO2: Explain the designing the open pit and layout of the haulage system and construction of bunker for storage of coal.
- ☛ CO3: Understand the knowledge on wining system & choosing of appropriate conveying system and its control in the mine.
- ☛ CO4: Apply the knowledge of construction & installation of aerial ropeway and man ridding system and its statutory provision.

Module 1

Construction and operation of blast hole drills, rippers, shovels, hydraulic excavators, scraper, dragline, dumpers, wheel loaders, dozers, graders, surface miners, BWE, spreader, stacker & reclaimer. High capacity belt conveyors—constructional detail and selection procedures. Aerial ropeways—classification, layout and constructional features. Classification, application and constructional features of crushers, breakers and feeders.

Module 2

Surface and Underground Layout Pit top and pit bottom circuits. Surface structures. Surface handling systems – coal and ore handling plants. Storage bunkers. Railway siding. Pit bottom layouts. Winding Drum and friction winding, headgears, headgear pulleys, cages and skips, suspension gear, keps and guides.

Module 3

Steam and electric winders, safety devices in winders, duty cycle. Automatic winding. Multilevel winding. Trackless Haulage Types of conveyors and their sequence control. High angle conveyor. Free steered vehicles - shuttle cars, LHD, SDL and low profile dump trucks (LPDT).

Module 4

Aerial Ropeways Types, construction and installation. Loading, unloading and angle stations, Man-riding Systems Statutory Provisions.

Suggested Reading:

1. *Voll-II, D.J. Deshmukh, Central Techno Publication, Nagpur*
2. *Mine pump, haulage, winding, Ghatak, Coal Field Publisher Asansol.*

Program: B.Tech

Semester: Seven

Course: Mine Legislation and Safety I

Course Code: 8PCCMiE404

Course Objective:

- ☛ The students will have knowledge on various acts, rules and regulations relating to the mineral industry.
- ☛ Student will also know about accidents, diseases and mine safety.
- ☛ To study various acts, rules and regulations relating to the mineral industry.
- ☛ Students will be acquainted with rules, laws and order for running a mine.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Understand the knowledge on implementation of provisions of mines act related to Mine, Open cast working, underground working, Relay, Shift, Serious bodily injury, Provisions for health and safety, Provisions regarding leave with wages.
- ☛ CO2: Apply knowledge of legislation in mines for the implementation of mine rules, various committee & provisions related to First Aid.
- ☛ CO3: Learn the various duties & responsibilities of mine managers, Asst/under Managers, Overman, foreman and surveyor.
- ☛ CO4: Explain the implementation of the safety management plan to reduce the accident and various type of mine accident.

Module 1

Mines Act 1952: Important definition: Adolescent, adult, child, Employed, Mine, Open cast working, Relay, Shift, Serious bodily injury. Provision for health and safety, Provisions regarding leave with wages, committees, Notice to be given of accident, Hours & Limitations of Employment. Mines rules: Provisions regarding health & sanitation, first aid and medical appliances & Mines Act in respect of Drinking water health and hygienic conservancy.

Module 2

Mines Rules 1955: Provision of workmen inspector & safety committee, Provisions related to health & sanitation and provisions related to First Aid & medical appliances, first aid room, first aid station and welfare amenities. Hours and limitations of Employment - leave with wages - with wages and over time.

Module 3

Coal Mines and Metalliferous Mines Regulations: Important definitions, regulations related to notice of accidents duties of managers, Asst/under Managers, Overman, foreman and surveyor, Mine plans and sections. Means of Access and egress ladder and Ladder ways under M.M.R. Transport of men and material by Haulage mine working, precautions against dangers from gas and water Mine ventilation, mine lighting and safety equipment and types of fences (Miscellaneous).

Module 4

Safety Aspects in Mines: Accidents classification and analysis-safe condition- unsafe condition- mine safety- safety objectives- major factors to be considered for safety - safety week- pit safety committee- safety organization and safety policy.



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Suggested Reading:

1. *Central Government, Mines Act 1952, Lovely Prakashan, Dhanbad*
2. *Central Government, Mines Rules 1955, Lovely Prakashan, Dhanbad*
3. *Central Government, Coal Mines Regulation 1957, Lovely Prakashan, Dhanbad*
4. *Central Government, DGMS Circulars, Lovely Prakashan, Dhanbad*

PROFESSIONAL ELECTIVE (ANY ONE OF THE FOLLOWING SUBJECTS)

Program: B.Tech

Semester: Seven

Course: Dimensional Stone Mining

Course Code: 8PECMiEEL401

Course Objective:

- ☛ The course is designed to help the student in understanding the different methods to mining and processing also extraction methods of dimensional stone.
- ☛ They can know about various properties of dimensional stone and various machines to use for extraction of dimensional stone.
- ☛ After learning this course the students should be able to understand the different method to mining and processing also extraction methods of dimensional stone.
- ☛ To know the hazard of dimension stone mining

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Apply knowledge on small scale & dimensional stone mining for understanding, formulating and solving problems related with small scale mining.
- ☛ CO2: Apply skill on selecting the equipment and cutting tools as per physical properties of material for designing and development of small scale mine.
- ☛ CO3: Apply knowledge of extraction of dimension stone.
- ☛ CO4: Apply the skill on preventive measures for the hazard caused due to dimension stone mine.

Module 1

Introduction: Definition, historical use of natural stones. Geology and occurrences: Classification of dimensional stones, composition, chemical and geo-chemical properties, various standards for normalization of dimensional stones. Mining of dimensional stones: Various techniques of dimensional stones mining—block mining and slab mining; Manual mining; Mechanized mining—line drilling, in-situ sawing by wire saw, chain saw, portable circular saw, flame cutting.

Module 2

Cutting/Sawing tools: Tool carrier—circular steel blade, steel wire rope, chain jib saw, physical and mechanical properties, elastic properties, tension etc.; Cutting tools— diamond segments, diamond pearls/bits, tungsten bits etc.; Process of manufacture, ingredients, brazing /fitting, wearing pattern and control; Cost of cutting. Handling of blocks and slabs: Equipment used derrick crane, front loaders, fork-lifts, mobile cranes, trucks and trailers.

Module 3

Quarrying machines for dimensional stones: Portable circular saw, wire saw, chain saw, line drills—special design features of the machines, techniques of use and maintenance. Production monitoring: Recovery, waste generation, productivity, inherent defects, measurement and corrective actions, cost evaluation. Environmental issues: Management of solid waste, slurry waste, soil and water; Protection and rehabilitation.

Module 4



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Health, safety and welfare: Protective care from abrasive dust, personal safety and welfare. Application, processing and architecture in dimensional stone: Application—flooring, roofing, cladding, stairs, paving, facets; Processing and polishing –various techniques for sawing of blocks, shaping of edges, polishing and calibration; Fixing and installation—techniques of fixing of dimensional stones in various applications like flooring, cladding, faces, stairs, roofing and paving; Care and maintenance of dimensional stones—techniques for post fixing care and maintenance of dimensional stones in various applications.

Suggested Reading:

1. *Hand Book On Safety In Mines & Dimension Stone Quarries, Mines Group Vocational Training Society, Hospet*
2. *A Text-Book Of Ore And Stone Mining (1894) By Sir Clement Le Neve Foster*

Program: B.Tech

Semester: Seven

Course: Rock Excavation Engineering

Course Code: 8PECMiEEL402

Course Objective:

- ☛ The students will get familiarity about rock mechanics properties, rock cutting technology and excavating machines.
- ☛ To understand the rock mechanics, rock cutting technology, rock cutting tools and rock excavating machine.
- ☛ Student get introduced with the rock mass classification and mechanism of rock reinforcement and the typical and special methods of rock reinforcement.
- ☛ After this module the student will be able to understand various rock cutting machine and their techniques

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Apply knowledge of rock mechanics in designing rock structures and excavation in civil and mining engineering projects.
- ☛ CO2: Apply knowledge of rock mechanics in drilling techniques, mechanism and its performance in underground and surface drilling
- ☛ CO3: Apply knowledge on fragmentation by explosives; advances in explosives and their selection criteria for rock excavation; blast design for surface excavations and optimization.
- ☛ CO4: Apply knowledge on interaction of rock and cutting tools as well as selecting different types of underground and opencast machines such as ploughs, shearers, road headers, continuous miners and tunnel boring machines ,shovel ,draglines etc

Module 1

Scope and importance: Rock excavation engineering in mining and construction industries; physico-mechanical and geotechnical properties of rocks Vis-à-vis excavation method; selection of excavation method.

Module 2

Drilling: Mechanics of rock drilling; design and operating parameters of surface and underground drilling; evaluation of drill performance; drill ability of rocks; mechanism of bit wear; bit selection; problems of drilling; economics of drilling.

Module 3

Blasting: mechanics of rock fragmentation by explosives; advances in explosives and their selection criteria for rock excavation; blast design for surface excavations and optimization; advanced blast initiation systems; blast performance evaluation; cast blasting; techno economic and safety aspects of surface and underground blasting; advances in blast design for underground excavations; contour blasting; computer aided blast designs; review of tunnel blasting techniques in recent advances.

Module 4

Rock Cutting: theories of rock tool interaction for surface excavation machinery- rippers, bucket wheel excavators, continuous surface miners; theories of rock tool interaction for underground excavation



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machinery- ploughs, shearers, road headers, continuous miners and tunnel boring machines; selection criteria for cutting tools; Advanced rock cutting techniques- high pressure water jet assisted cutting.

Suggested Reading:

1. *Theory and Technology of Rock Excavation*, Zou, Dingxiang
2. *Rock Fragmentation by Blasting* by B Mohanty
3. *Principles of Rock Drilling* by U M Rao Karanam and B Mishra
4. *Rock Blasting: Effects and Operations* by Pijush Pal Roy

Program: B.Tech

Semester: Seven

Course: Design of Open Pit Mines

Course Code: 8PECMiEEL403

Course Objective:

- ☛ This course discusses the geological model and its purpose of linking the regional physical geology and the events that lead to the formation of the ore body to a mine-scale description of the setting.
- ☛ Discusses the distribution and nature of the overburden soils and rock types at the site, including the effects of alteration and weathering.
- ☛ Aware about the new techniques to design open cast mines and analyze the surface mining parameters.
- ☛ Learn about the different hazard in open pit

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO1: Explain the knowledge of planning process of open pit mine& reserve estimation.
- ☛ CO2: Analyze the pit design and synthesis of short term planning and to control and influence production scheduling processes.
- ☛ CO3: Understand the development of infrastructural facilities such as communication, power supply, beneficiation plants and mineral handling plants etc. and able to design various kind of open pit working.
- ☛ CO4: Apply knowledge on Controlling of opencast mining induced hazards- rock fall, fly rock, blast vibration, noise and mine dusts.

Module 1

Preliminary evaluation of deposits, different stages of exploration, drilling for grade and geo-technical information. Calculation of reserves, Collection of planning information.

Module 2

Optimal size and output different stages of development, design of pit layout and method of advance. Selection of open cast equipments and equipment scheduling, Design for unit operations, rock breaking, excavation, transportation, storage and dumping.

Module 3

Development of infrastructural facilities-communication power supplies, illumination, dewatering and other maintenance services, Site selection for mineral handling/ beneficiation plant.

Module 4

Control of opencast mining induced hazards- rock fall, fly rock, blast vibration, noise and mine dusts. Planning for reclamation of mined out area.

Suggested Reading:

1. *Open Pit Mine Planning and Design*, William A. Hustrulid, Mark Kuchta
2. *Open Pit Mine Planning and Design*, William A

Program: B.Tech

Semester: Seven

Course: Vocational Trainee Report II

Course Code: 8PROJMiE401

Course Objective:

- ☛ To provide training in mines for gaining thorough understanding of all the theoretical knowledge.
- ☛ Gaining practical experience is an important aspect of the mining engineering programme having many characteristic features of its own.
- ☛ The students will have insight about mining methods and techniques.
- ☛ The outcome at the place of work is always much more than what can be learned in the class room.
- ☛ To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.
- ☛ To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ Mining graduates would be equipped with managerial skill which would be useful to them for achieving their program educational outputs.

1. Study of History of Mine – Note name of the Owner, Agent, Manager, Safety Officer
2. Study of Mine geological information
3. Study of Mine Plans and Sections
4. Study of Surface features related to Mine
5. Study of method of working
6. Study of method of blasting
7. Study of Transportation system and layouts
8. Study of Ventilation systems and layouts
9. Study of Drainage system
10. Study of Pit top and Pit bottom layouts.
11. Study of man Power plan
12. Develop the Lamp room layout and Magazine Layout
13. Draw the charts depicting instructional items related to Mining subjects

Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of Semesters.

Program: B.Tech

Semester: Seven

Course: Seminar in Executive Communication

Course Code: HSMC401

Course Objective:

The objectives of the course are:

- ☛ To develop a general knowledge of the basic theories of human communication in rhetorical, group, and interpersonal settings.
- ☛ To develop the basic understanding of the principles and techniques of persuasion in interpersonal, group, and public speaking contexts.
- ☛ To develop an ability to successfully apply the above knowledge in actual small group, interviewing, business, public speaking, and interpersonal situations. The ability to write well-worded and persuasive resumes and other business communication. Also, the ability to write a valid and well-supported analysis of communication problems encountered in real-world situations.
- ☛ To develop an understanding of how the perception of both verbal and non-verbal messages influences culture, behavior, and action of life itself.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Develop an understanding of the process of oral communication
- ☛ CO 2: Develop critical thinking and analytical skills.
- ☛ CO 3: Improve listening, note taking and observational skills
- ☛ CO 4: Become more knowledgeable about audience centered speaking
- ☛ CO 5: Develop message generating and delivery skills
- ☛ CO 6: Become more knowledgeable about current speaking strategies and practices and new media research strategies

(Activity Based)

WORKSHOPS

- ☛ Debate
- ☛ Extempore
- ☛ Group Discussion
- ☛ Panel Discussion
- ☛ Presentation-Paper & Oral
- ☛ Reports: Survey Report, Project Report, Case Study

Suggested Readings:

1. Monipally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen, Leena. *Communication Skills*; Eastern Economy Edition
5. David Green. *Contemporary English Grammar, Structure & Composition*; MacMillan
6. *Dictionary*; Oxford
7. *Dictionary*; Longman



RKDF UNIVERSITY RANCHI

MINING ENGINEERING

B.Tech



RANCHI

SYLLABUS

(2022-2026)

SEMESTER VIII



RKDF UNIVERSITY RANCHI

B.TECH SEMESTER VIII

BATCH 2021-2025					
BTECH IN MINING ENGINEERING					
SEMESTER VIII					
S.No.	CATEGORY	CODE	COURSE TITLE	Subject Total	Credit
1	Professional Core Courses	8PCCMiE405	Operational Research in Mining	100	3
2	Professional Core Courses	8PCCMiE406	Mine Legislation & Safety II	100	3
3	Professional Core Courses	8PCCMiE407	Mine Management	100	3
4	Professional Core Courses	8PCCMiE408	Fuel Technology & Mineral Processing	100	3
5	Professional Core Courses	8PCCMiE409	Mine & Mineral Economics	100	3
6	OPEN ELECTIVE (Any one of the following subjects)				
	Open Elective Courses	8OEEL401	Engineering Economics	100	3
	Open Elective Courses	8OEEL402	Industrial Management		
	Open Elective Courses	8OEEL403	Coal Bed Methane Mining		
7	Humanities and Social Sciences	HSMC402	**Human Values & Ethics	100	0
PRACTICAL / SESSIONAL					
1	Professional Core Courses	8PCCMiE408P	Fuel Technology & Mineral Processing Lab	50	1
2	Project	8PROJMiE402	Mining Project Work	200	5
			TOTAL	950	24
			Total of All Semester	6350	162
**NOTE: Qualifying Non Credit Course.					
A student will be eligible to get Under Graduate degree with Honours , if he/she acquires an additional 20 credits through MOOCs.					

Program: B.Tech

Semester: Eight

Course: Operational Research in Mining

Course Code: 8PCCMiE405

Course Objective:

- ☛ The students will learn the concept of system engineering and applicability in mining field.
- ☛ To study the various techniques of operations research, simulation and network analysis.
- ☛ Enables the students to know basic of system engineering concept and analysis.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Formulate real-world problems as a linear programming model and describe the theoretical workings of the graphical and simplex method.
- ☛ CO 2: Formulate specialized linear programming problems, namely transportation and assignment problems and describe theoretical workings of the solution methods for transportation and assignment problems.
- ☛ CO 3: To develop the abilities in project evaluation techniques like PERT, CPM etc.
- ☛ CO 4: Apply the knowledge of game theory concepts to articulate real-world decision situations for identifying, analyzing, and practicing strategic decisions to counter the consequences.

Module I

Linear Programming :Linear Programming models; Assumption of linear programming, Graphical and Simple method of solving Linear Programming Problems; Basic and Basic feasible solution, optimal solution, interpretation of SIMPLEX table. Primal and Dual Problem. Application of Linear Programming for solution of mining related problems of production planning, scheduling and blending.

Module 2

Transportation and Assignment Problem: Transportation models, Solution. Algorithm for Transportation problem. Assignment model, Variations on Classical Assignment model; solution algorithm for Assignment problems. Mathematical formulation of an assignment problems. Difference between transportation and assignment problems.

Module 3

Project Management with PERT &CPM: Assumption of PERT and CPM; Methods of drawing network; Rules of network construction, numbering the events, construction of network, Time analysis ; Forward pass computation; backward pass computation; determination of floats and slack times; Critical path calculation.

Module 4

Game Theory: Definition; pay off, types of games, the Maximin- Minimax principle, games without saddle point, graphical method, dominance property.

Suggested Reading:

1. *Numerical Methods in Engineering and Science* by Grewal B.S. and Grewal J.S
2. *Optimization: Linear Programming* by B K Mishra and B N Mishra
3. *Linear Programming: Foundations and Extensions* by Robert J Vanderbei

Program: B.Tech

Semester: Eight

Course: Mine Legislation and Safety II

Course Code: 8PCCMiE406

Course Objective:

- ☛ Enables the students to understand the organizational structure, working safety and management concepts for mining enterprises to effectively manage them within the frame work of rules and regulations.
- ☛ Students will be acquainted with rules, laws and order for running a mine.
- ☛ The students will have knowledge on various acts, rules and regulations relating to the mineral industry.
- ☛ They will also know about accidents, diseases and mine safety.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Apply knowledge of legislation in mines for the implementation of rules and regulations during their job.
- ☛ CO 2: Define work effectively with other engineering and science teams for suggesting any measures against any mine accidents.
- ☛ CO 3: State application of various acts , rules and regulation
- ☛ CO4: Interpret the cause and remedies of various accident disease and mine safety.

Module 1

Coal Mines Regulations 2017: Important definitions: Duties and responsibilities of workmen, competent person & officials. Planes and sections. Means of access & egress, Provisions regarding winding in shaft, Transport of men & material, mine working, Precautions against dangers from the dust, gas & water and Ventilation. Provisions regarding lighting and safety lamp, Explosives & Blasting, Provisions regarding machinery, plant & equipments and important provisions under chapter on miscellaneous.

Metalliferous Mines Regulations 1961: Plans and sections, Means of egress, Ladder & Ladder way and transport of men & material.

Module 2

Important provisions of **Mines Rescue Rules 1985**, Provisions of **Electricity Rules 1961** relevant to Mining and salient provisions of the **Mines and Minerals (Development & Regulation) Act 1957**, **Mineral Concession Rules 1960** and **Industrial Disputes Act 1947**.

Module 3

Mine Safety: Occupational hazards of mining; Accidents and their classification; Statistics of fatal and serious accidents; Frequency and severity rates of accidents; Place-wise and Cause-wise analysis.

Module 4

Mine Safety: Basic causes of accident occurrence; Investigations into accidents and accident reports; Measures for improving safety in mines; Cost of accidents. Introduction to safety management system and risk assessment.

Suggested Reading:

- 1. Central Government, Mines Act 1952, Lovely Prakashan, Dhanbad*
- 2. Central Government, Mines Rules 1955, Lovely Prakashan, Dhanbad*
- 3. Central Government, Coal Mines Regulation 1957, Lovely Prakashan, Dhanbad*
- 4. Central Government, DGMS Circulars, Lovely Prakashan, Dhanbad*
- 5. Ghatak, A Study of Mine Management Legislation and General Safety, Shining Printers, Asansol*

Program: B.Tech

Semester: Eight

Course: Mine Management

Course Code: 8PCCMiE407

Course Objective:

- ☛ Students gain knowledge on the basic management principles to become management(s) professional.
- ☛ Students will be acquainted with rules, laws and order for running a mine.
- ☛ At the end of the course, the student should be able to the students would be able to understand the basic application of operational tools and manufacturing.
- ☛ **Develop leadership and a good communication skill**

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Summarize the evolution theory of management and sketch the structure & design of organization.
- ☛ CO 2: Acquire knowledge and hands-on competence in applying the concepts of management in the development of mine economics.
- ☛ CO 3: Analyze and solve financial management problems.
- ☛ CO 4: Illustrate leadership skills and be able to be knowledgeable in behavioral science

Module 1

Introduction: Evolution of management theory and practice; Principles of scientific management; Structure and design of organization for mining enterprises.

Module 2

Personnel management: Selection, Training and Development of human resources for mining enterprises. Industrial Relations: Concepts and relevant regulations.

Module 3

Finance management: Capital budgeting techniques for mine projects; Methods of cost analysis and cost control, Break-even analysis.

Module 4

Behavioral sciences for management: Conflicts in organization; Motivation; Leadership; Communication.

Suggested Reading:

1. *Mine Management Legislation (up to 2004)[Vol.I to VII] (Coal) by H P Soni*
2. *Hand Book of Mine Management Legislation & General Safety by R K Raman*

Program: B.Tech

Semester: Eight

Course: Fuel Technology and Mineral Processing

Course Code: 8PCCMiE408

Course Objective:

- ☛ The course enables the students to select the suitable parameters and appropriate machinery for processing various types of minerals.
- ☛ Give the students an understanding of how the basic mineral processing unit operation can be combined into specific processes.
- ☛ Students will be acquainted with mineral separation methods – an overview, including: physical separation (gravity, magnetic and electrical); solid/liquid separation; flotation.
- ☛ To understand the leaching process & sampling processes.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Apply knowledge of mineral dressing for understanding, formulating and solving problems related with mineral dressing.
- ☛ CO 2: Acquire knowledge and hands-on competence in applying the concepts in the design and development of machines for separating the low grade ore economically.
- ☛ CO 3: Design & understand the various methods for mineral beneficiation.
- ☛ CO 4: Demonstrate the knowledge on leaching and laboratory sampling process.

Module 1

Fuel Technology & Solid fuels: Wood, peat, lignite, coal, anthracite; proximate and ultimate analysis; coal characteristics for different industrial uses; characteristics of Indian coals; caking and coking properties; low and high temperature carbonization. Liquid fuels: Petroleum its products and testing methods. Gaseous fuels: Natural gas, producer gas and water gas. Combustion: Combustion stoichiometry; Coal burning equipment.

Module 2

Mineral Processing & Introduction: Scope, objectives and limitations of mineral processing; Liberation and beneficiation characteristics of minerals and coal. Comminution: Theory and practice of crushing and grinding; Different types of crushing and grinding equipment, their application and limitations. Size separation: Laboratory size analysis and interpretation; Settling of solids in fluids; Industrial screens; Mechanical classifiers and hydrocyclones.

Module 3

Gravity concentration methods: Jigging, heavy media separation, flowing film concentrator's, theory, application and limitations. Froth flotation: Physico-chemical principles; Reagents; Machines; Flotation of sulphides, oxides and coal. Electrical and magnetic methods of concentration: Principles, fields of application and limitations.

Module 4

Dewatering: Thickener and filter. Hydro-metallurgical methods of recovery: Leaching- principle, various methods and applications. Laboratory sampling, Simplified flow sheets for coal, copper, lead, zinc, gold, iron, manganese ores and lime stone.



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Suggested Reading:

1. *Mineral Processing Technology* by B A Wills
2. *Introduction to Mineral Processing* by E G Kelly and D J Spottiswood
3. *Mineral Processing* by S K Jain



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Program: B.Tech

Semester: Eight

Course: Fuel Technology and Mineral Processing Lab

Course Code: 8PCCMiE405

List of Experiments:

1. Proximity analysis of coal.
2. Caking index of coal.
3. Crushing by Jaw roll crusher.
4. Grading of iron ore fines.
5. Floating of coal fines.

Program: B.Tech

Semester: Eight

Course: Mine and Mineral Economics

Course Code: 8PCCMiE409

Course Objective:

- ☛ The course enables the students to understand the economics of business enterprise to become a successful manager.
- ☛ Students gain knowledge on the basic management principles to become management(s) professional.
- ☛ Upon completion of the course, students will be able to gain knowledge and skills needed to run a business successfully.
- ☛ Expertise the students to brings employment, government revenues, and opportunities for economic growth and diversification through mining.
- ☛ Study of estimation and valuation of mineral deposit and study of project appraisal.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Apply knowledge of mine economics for understanding mineral related policies and solve problems related with the mine economics.
- ☛ CO 2: Analyze and solve the concepts of mine economics, its risk and mineral inventory.
- ☛ CO 3: Describe the concept of mine sampling & illustrate the valuation of mineral deposit and be able to estimate the mine life and profitability.
- ☛ CO 4: Discuss the forms of business, royalty, taxes & duties in mining sector.

Module 1

Mine Economics General: Economic importance of the mineral industry; Risky nature of the mining industry; National mineral policy; Mineral price and pricing; Conservation of mineral resources. Mine Sampling: Definition, purpose and scope, methods and computations; Reliability of mine sampling. Loss of mineral in mining: Classification and estimation; Dilution and recovery .Mine examination and valuation.

Module 2

Mineral Economics Introduction: Economic importance of the mineral industry; Mining economy, risky nature of the mining industry; State and the mining industry; National Mineral Policy. Mineral resource: concept, classification and estimation. Mineral inventory: concept, characteristic features, composition and economic significance; Estimation of life index. Economics of mineral exploration and production. Mineral price and pricing, price index .Mineral consumption and substitution; Market survey and demand analysis. Market structure, market conduct and performance. Conservation of mineral resources– scope and limitations.

Module 3

Mine sampling: Definition, purpose and scope; Sampling methods and computations; Reliability of mine sampling. Loss of mineral in mining: Classification and incorporation of losses; Coefficient of completeness of mineral extraction; Dilution and recovery. Mine examination and valuation: Examination and valuation of mines/mineral properties; present value and its computation.

Module 4

Forms of business organization: Private and public enterprises. Mine finance: Capital– its importance, various forms and formation; Raising capital. Mine accountancy and book keeping. Royalty, taxes and duties; Imports and exports. International investment and trade in mineral materials & products.



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Suggested Reading:

1. *Mine Valuation, Sparks,*
2. *Mine Sampling and Valuation, Sulchre*
3. *Mine Geology, Arogyaswami*

OPEN ELECTIVE (ANY ONE OF THE FOLLOWING SUBJECTS)

Program: B.Tech

Semester: Eight

Course: Engineering Economics

Course Code: 8OEEL401

Course Objective:

- ☛ Enable students to perform and evaluate present worth, future worth and annual worth analyses on one of more economic alternatives.
- ☛ Students will have knowledge of liability, insurance and safety/environmental requirements on projects.
- ☛ Prepare engineering students to analyze cost/revenue data and carry out make economic analyses in the decision making process to justify or reject.
- ☛ The objectives of this course are to review with students basic economic principles with students.
- ☛ Educate the students on how to systematically evaluate the various cost elements of a typical manufactured product, an engineering project or service, with a view to determining the price offer.

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Describe the concept of demand & supply and its managerial application.
- ☛ CO 2: Estimate the Present, annual and future worth comparisons for cash flows.
- ☛ CO 3: Calculate the capital investment, net present value & internal rate of return.
- ☛ CO 4: Account & analyses the sustainability, income determination and fluctuations in economy.

Model 1

Introduction, demand and supply, statistical demand and supply analysis, managerial application of the elasticity of demand and supply.

Model 2

Estimation of production and cost function; forecasting techniques and their use for decision making at different levels; linear programming and production analysis. Price output determination: a review with special reference to condition of uncertainty, kinds of pricing problems.

Model 3

Capital market and investment decisions: pay back method, average return on investment or accounting rate of return, net present value, and internal rate of return.

Model 4

Outline of welfare economics, resource accounting and sustainability, income determination and fluctuations.

Suggested Reading:

1. *Engineering Economics* by Dr. R.K. Singal & Er. Rishi Singal
2. *Engineering Economics* by Panneerselvam

Program: B.Tech

Semester: Eight

Course: Industrial Management

Course Code: 8OEEL402

Course Objective:

- ☛ To make the students aware of the outline of managerial functions relating to manufacturing.
- ☛ At the end of the course, the student should be able to understand the basic application of operational tools and manufacturing.
- ☛ Upon completion of the course, students will be able to gain knowledge and skills needed to run a business successfully.
- ☛ Learn various aspect of reliability

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Apply management practices & & practice the various management theory.
- ☛ CO 2: Describe the effectively the formation of companies, financial analysis and costing, & operation Research.
- ☛ CO 3: Identify strategies for deployment and management of human resources on organization.
- ☛ CO 4: Understand Project management & assurance Sciences.

Model 1

Management functions, Evolution of management theory and practice, organization.

Model 2

Formation of companies, financial analysis and costing, engineering economy, work study. Elements of Operation Research- PERT, CPM, LPP, Scheduling and Sequencing.

Model 3

Deployment and management of human resources, Product engineering, Plant location and layout, production planning and inventory control.

Model 4

Project management, Assurance Sciences – reliability, quality and maintainability.

Suggested Reading:

1. [*Industrial Management*](#) by O.N.Pandey
2. [*Industrial Management*](#) by K. Patrai
3. A [*Text Book of Industrial Management*](#) by A.P.Verma and N.Mohan

Program: B.Tech

Semester: Eight

Course: Coal Bed Methane Mining

Course Code: 8OEEL403

Course Objective:

- ☛ After completion of this course student will be aware of alternate fossil fuels, their production methods.
- ☛ Upon completion of this course, the students can able to identify the new methodologies / technologies for effective utilization of renewable energy sources

Course Outcome:

After the completion of the course, student will be able to:

- ☛ CO 1: Explain technology & process for coal bed methane exploration.
- ☛ CO 2: Analyze the CBM reservoir & methane estimation.
- ☛ CO 2: Understand the concept of hydraulic fracturing in coal bed.
- ☛ CO 3: Interpret the method of compression & transportation in CBM.

Model 1

Coalification process and coal grades. Methane generation and storage in coal; Geological control in Coal Bed Methane (CBM) exploration; Methane adsorption, desorption in coal.

Model 2

Coal as CBM reservoir: In-place methane estimation; Transport of methane in coal-bed. Drilling & Completion of a CBM hole/well. Identification and characterization of coal beds by hole/ well logs.

Model 3

Hydraulic fracturing in coal beds. Production performance of a CBM hole/well; Water drainage & gas-water separation. Gas volume measurement.

Model 4

Compression & transport; Liquefaction and utilization. Enhanced recovery by CO₂ and N₂ adsorption methods.

Suggested Reading:

1. *Progressive Technologies of Coal, Coal bed Methane, and Ores Mining* by Volodymyr Bondarenko, Iryna Kovalevs'ka
2. *Advanced Reservoir and Production Engineering for Coal Bed Methane* by Pramod Thakur
3. *Fundamentals of Coal bed Methane Reservoir Engineering* by John Seidle

Program: B.Tech

Semester: Eight

Course: Mining project Work

Course Code: 8PROJMiE402

Course Objective:

- ☛ To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.
- ☛ The aim of the project is to stimulate creative and innovative aspects of their technological learning.
- ☛ The aim of the project work is to deepen comprehension of principles by applying them to a new problem which may be the design and manufacture of a device, a research investigation, a computer or management project or a design problem.

Course Outcome:

- ☛ CO 1: Mining graduates would be equipped with managerial skill which would be useful to them for achieving their program educational outputs.
- ☛ CO 2: Explain the mining practices needed for current Indian coal mining scenario for sustainable development to meet organizational objective.
- ☛ CO 3: Contribute towards improvement in production and productivity including optimization of manpower in mines.
- ☛ CO 4: Detailed understanding of the scientific principles associated with various coal beneficiation techniques.
- ☛ CO 5: Develop skill for time bound projects implementation and production commissioning with appropriate technology adoption for economic mining & effective coal beneficiation.

Any project /research work related to mining area.

Program: B.Tech

Semester: Eight

Course: Human Values and Ethics

Course Code: 40B.451

Course Objective:

The objectives of the course are:

- ☛ To provide a critical perspective on the socialization of human beings.
- ☛ To introduce students about some key aspects of Indian culture and ethics.
- ☛ To help students reflect critically on gender violence.
- ☛ To expose the students to more egalitarian interactions between men and women.
- ☛ To develop students sensibility with regard to issues of gender in contemporary India.
- ☛ To understand the moral values that ought to guide the Management profession.
- ☛ To justify the moral judgment concerning the profession.
- ☛ Intended to develop a set of beliefs, attitudes, and habits that engineers should display concerning morality.
- ☛ To create an awareness on Management Ethics and Human Values.
- ☛ To inspire Moral and Social Values and Loyalty.

Course Outcome:

At the end of the course learners will be able to:

- ☛ CO 1: To attain a finer grasp of how gender discrimination works in our society and how to counter it.
- ☛ CO 2: Understand the ideas of values, ethics, and morality in a multicultural context.
- ☛ CO 3: Understand how universal values can be uncovered by different means, including scientific investigation, historical research, or public debate and deliberation (what some philosophers call a dialectic method)
- ☛ CO 4: Understand and discuss the idea of moral relativism and the challenges it poses to universal values
- ☛ CO 5: Critically assess the relationship between theory and practice in the formulation of values.
- ☛ CO 6: Understand that values arise from lived experiences, but need to be justified to others.
- ☛ CO 7: Understand the role of deliberation and debate in framing such values.
- ☛ CO 8: Understand how to create an actionable document through such a process.
- ☛ CO 9: Understand the importance of human solidarity and dignity.

1. VALUE CRISIS IN CONTEMPORARY INDIAN SOCIETY

- 1.1 Value Crisis at the Individual Level
- 1.2 Societal Level
- 1.3 Intellectual Level
- 1.4 Cultural Level
- 1.5 Value – What are they?
- 1.6 The Indian Concept of Values.
- 1.7 Modern Approach to the Study of Values.
- 1.8 Aesthetic Sensibilities

2. MORAL AND ETHICAL HUMAN VALUES

- 2.1 Bases for Moral Judgment
- 2.2 Some Canons of Ethics.
- 2.3 Virtue Ethics.
- 2.4 Ethics of Duty.
- 2.5 Ethics of Responsibility
- 2.6 Factors to be considered in Making Ethical Judgments.
- 2.7 Different Meanings of Human Values
- 2.8 A New Approach to Human Value ,Freedom, Creativity Love &Wisdom

3. MORAL VALUES IN PROFESSION

- 3.1 What is a Profession?
- 3.2 Professional Ethos
- 3.3 Code of Professional Ethics
- 3.4 Practicing the Code
- 3.5 Corporate Social Responsibility
- 3.6 The Larger Domain of Human Values
- 3.7 Institutionalizing Ethics and Human Values

4. GENDER SENSITIZATION

- 4.1 Socialization of women
- 4.2 Just Relationships, being together as equals
- 4.3 Declining sex ratio, demographic consequences
- 4.4 Women's work, its politics and economics, fact and fiction, unrecognized and Unaccounted work
- 4.5 Domestic violence, eve teasing and harassment. Is home a safe place?

Suggested Reading:

- 📖 Dr. Rajan Mishra, *Human Values: Laxmi Publications Pvt. Ltd.*
- 📖 S. Dinesh Babu, *Professional Ethics and Human Values; Laxmi Publications Pvt. Ltd.*
- 📖 P.S. Rathore. *Business Ethics And Communication; S.Chand Publishing*
- 📖 Dr. K.Alex. *Managerial Skills; S. Chand Publishing.*
- 📖 Dr. M. Adithan, *Study Skills For Professional Students For Higher Education , S.Chand Publishing*
- 📖 Govindarajan M “*Professional Ethics and Human Values.*”
- 📖 R.R. Gaur and R. Sangal “ *A Foundation Course in Human Values and Professional Ethics*”

Websites:



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www.schandedutech.com

www.laxmipublications.com